

Lec, 1. Plant diseases-Abiotic, Biotic, Classification based on Mode of infection, Multiplication of Inoculum, Spread, Symptoms, Occurrence and Distribution.

Types of causal agents

Classification of plant diseases will be meaningful and useful if they are classified on the basis of type of pathogenic or non-pathogenic causes of the disease. Such kind of classification with details on the development and spread of the disease and management methods is considered best. The classification based on major causal agents is as follows.

A. Parasites: They include both biotic and mesobiotic agents. The diseases are incited by parasites under a set of suitable environment. Association of definite pathogen is essential with each disease.

i. Biotic agents: They are also called as animate causes. They are living organisms. Biotic agents include.

1. Prokaryotes

- a. **True bacteria** or bacteria (Facultative parasites) e.g. Citrus canker.
- b. **Fastidious Vascular Bacteria** (FVB) e.g. Citrus greening, Pierce's disease of grape
- c. **Mollicutes or wall-less prokaryotes**
 - I. **Phytoplasma** e.g. Sesame phyllody, Brinjal little leaf.
 - II. **Spiroplasma** e.g. Corn stunt, Citrus stubborn

2. Eukaryotes

Protists (Unicellular, coenocytic or multicellular with little or no differentiation of cells and tissues).

- ✓ **Fungi** e.g. Wilt of cotton
- ✓ **Protozoa** e.g. Hart rot of coconut
- ✓ **Algae** e.g. Red rust of mango
- ✓ **Plants** - Parasitic flowering plants or **phanerogamic parasites** - Broomrape of tobacco.
- ✓ **Animals** (Multicellular, extensive differentiation of cells and tissues) e.g. **Nematodes** - Root knot nematode

ii. Mesobiotic agents: They include viruses and viroids. They are infectious agents. They can be crystallized and are considered non-living. But their multiplication in the living plants ensures that they are living. Hence they are called as mesobiotic agents.

1. **Viruses** e.g. yellow mosaic of blackgram
2. **Viroids** e.g. spindle tuber of potato

B. Non-parasites or Abiotic agents: They are also called as **non-infectious** or **physiological disorders**. When no pathogen is found, cultured from or transmitted from a diseased plant, then the disease is said to be caused by a non-living or environmental factor. These diseases occur because of disturbances in the plant system by the improper environmental conditions in the air or soil or by mechanical influences. They are listed below.

- i. Too low or too high temperature
- ii. Lack or excess of soil moisture
- iii. Lack or excess of light
- iv. Lack of oxygen
- v. Air pollution (Toxic gases in the atmosphere etc.)
- vi. Mineral deficiencies or toxicities
- vii. Soil acidity or alkalinity
- viii. Toxicity of pesticides
- ix. Improper agricultural practices.

Classification of plant diseases

There are thousands of diseases, which attack crop plants. Classification can be made based on several criteria. The various ways of classifying diseases of plants are given below.

1. Type of infection

- a. Localized diseases:** These diseases are limited to a definite area of an organ or part(s) of a plant. E.g. leaf spots and anthracnose caused by different fungi.
- b. Systemic diseases:** In these diseases the pathogen spreads from a single infection point so as to infect all or most of the host tissues. E.g. Downy mildews caused by fungi and mosaics and leaf curls caused by viruses.

2. Type of perpetuation and spread

- a. Soil-borne diseases:** The causal agents perpetuate and spread through soil. e.g. Damping off caused by fungi like *Pythium* sp. and root rot caused by *Rhizoctonia* spp.
- b. Seed-borne diseases:** Seed or seed materials help in the perpetuation and spread of this disease. The disease causing agents may be internally seed-borne or externally seed-borne e.g. Loose smut of wheat caused by *Ustilago nuda tritici* (internally seed-borne) and blast of rice caused by *Pyricularia oryzae* (externally seed-borne).
- c. Air-borne diseases:** In these type of diseases the causal agents are spread by wind (air). e.g. Early leaf spot and late leaf spot of groundnut caused by *Cercospora arachidicola* and *Phaeoisariopsis personata* respectively.

3. Extent of occurrence and geographic distribution

a. Endemic diseases: It is also known as enphytotic disease. When a disease is more or less constantly occurring year after year in a moderate to severe form in a country or locality then it is called as an endemic disease. e.g. Wart disease of potato caused by *Synchytrium endobioticum* is endemic in Darjeeling, citrus canker caused by *Xanthomonas campestris* pv. *citri* is endemic in Asia and sorghum rust caused by *Puccinia purpurea* is endemic in India.

b. Epidemic or epiphytotic diseases: An epidemic or epiphytotic refers to sudden outbreak of a disease periodically over a widespread area in a devastatingly severe form causing extensive losses or complete destruction. Epidemic disease may be present constantly in the locality but assumes a severe form only on occasions. This is because of the occurrence of favourable environment responsible for the rapid development of the disease. But the pathogen may be irregular in appearance or there may be lack of sufficient inoculum to cause the disease. e.g. wheat stem rust (*Puccinia graminis tritici*) and powdery mildew (*Erysiphe graminis* var. *tritici*), late blight of potato (*Phytophthora infestans*), sugarcane red rot (*Phylospora tucumanensis*), downy mildew of grapevine (*Plasmopara viticola*) and rice blast (*Pyricularia oryzae*).

c. Sporadic diseases: Sporadic diseases are those, which occur at irregular intervals over limited areas or locations. They occur in relatively few instances. e.g. *Fusarium* wilt of cotton (*Fusarium oxysporum* f.sp. *vasinfectum*), grain smut of sorghum (*Sporisorium* = *Sphacelotheca cruenta*) and wheat loose smut (*Ustilago nuda tritici*).

d. Pandemic diseases: A disease is said to be pandemic when it is prevalent throughout the country, continent or world involving mass mortality. e.g. Late blight of potato and wheat stem rust.

4. Multiplication of inoculum

a. In simple interest disease (monocyclic epidemics) the disease increase is just like simple interest in money. Here inoculum comes from a reservoir and hence amount of inoculum for a given season's crop is fixed. So there is no repetition of the disease cycle within the crop season. Hence the disease spread will be slow. e.g. Soil inhabiting pathogens like *Pythium* sp., *Rhizoctonia* sp. and *Sclerotium* sp.

b. In compound interest disease (polycyclic epidemics) the disease increase is just like compound interest in money. Inoculum is multiplied several times (every 7 to 15 days for wheat rust) during crop growth in a season. So the disease spread will be fast. e.g. Wheat stem rust, rice blast, powdery mildew diseases of different crops.

Lec. 2. Epidemiology- diseases Triangle/Diseases pyramid- Role weather factors in Plant Diseases development. Boom and Bust cycle in diseases development.

Epidemiology

Epidemiology is a study of outbreak and spread of diseases in a population. Epiphytology or epidemiology of plant diseases is essentially a study of the rate of multiplication of a pathogen which determines its capacity to spread a disease in a plant population.

The buildup of an epiphytotic is actually the manifestation of a balance between two opposing forces. On one side are the growth character and pathogenic potentialities of the pathogen and on the other side are the forces that counteract these capabilities of the pathogen. Environment plays a very decisive role in epiphytotics.

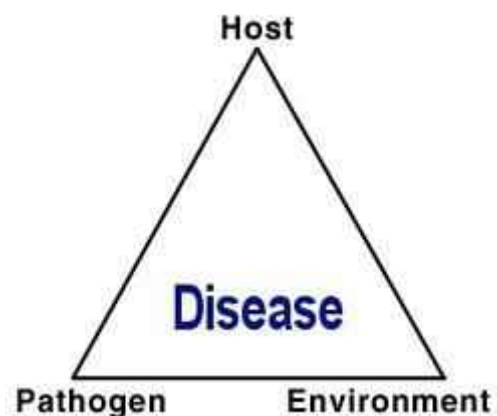
The Disease Triangle

Three factors must be present at the same time for a plant disease to occur. If any one of the three is missing, plant disease does not occur.

The factors are:

- A susceptible host
- A virulent pathogen (one that can cause disease), and
- A suitable environment

Plant pathologists refer to this as the **disease triangle**. You can manage elements of the disease triangle to control disease. For example, you can manage the host factor by growing some *Rhododendron* varieties that are genetically resistant to *Phytophthora*. You can manage the environment by draining excess water to make the area unsuitable for *Phytophthora* diseases. Or, you can manage the pathogen by treating water to kill *Phytophthora*.



In this module, you will learn more about ways to modify the elements in the disease triangle to control *Phytophthora* diseases in nurseries.

1. The host:

In our production system, regardless of planted crop (whether corn, cotton, grain sorghum, rice, soybean, wheat) more often than not, but not in every field situation, a susceptible cultivar is planted that does not contain resistance genes for a particular disease. Excluding rusts (common rust of corn, southern rust of corn, soybean rust of soybean,

peanut rust) in this particular section since they have to blow into the MS production system from somewhere to our south and ALL corn and soybean varieties are susceptible to rusts. Wheat rusts are similar in that they have to blow into MS; however, tolerant varieties exist. However, for some foliar diseases, specifically in corn and soybean, there are plenty of varieties available with some level of tolerance to a particular pathogen/disease (e.g., frogeye leaf spot).

2. The pathogen:

The pathogen is simply any particular organism (bacterial, fungal, or viral), but in this particular situation I will continue to refer to the fungal organisms, that can cause a disease in a particular crop plant. In some cases the pathogen is present at a particular location especially if a continuous crop has been grown. Most fungal organisms can overwinter in a field between seasons on senesced plant material or even in the soil profile. The fact that fungi can overwinter within a particular field is one major reason that rotation as well as planting a resistant variety is such an important consideration. Again, in this particular situation, rusts are the exception to this particular rule.

3. The environment:

The environment is probably the single most delimiting factor for the development of a plant disease situation. Hence the side of the disease triangle for the environmental factor being skewed. Without a conducive environment, a plant disease will not occur. In most cases, a conducive environment involves high humidity, or free moisture on the leaf surface in the form of dew as well as moderate temperatures. Most fungi do not like the temperature extremes. However, the environment must remain conducive for a specific amount of time for disease to occur. Time is the key component lacking in the three-dimensional disease triangle and the reason for the additional fourth component since a prolonged period of exposure between the host and the pathogen in a conducive environment will be required prior to disease incidence and symptom expression.

The disease triangle illustrating the phenomenon of plant disease as the interior space of a triangle with the three essential factors (susceptible host, favourable environment for disease, and pathogen) at the vertices. Variation in the 'strength' of the contributions of these factors to the relationship ('strength' is indicated by the size of the circles) will quantitatively alter the severity of the disease, which will be shown by a change in the area of the central disease envelope. The diagram is intended to be used dynamically; the static disease triangle allows

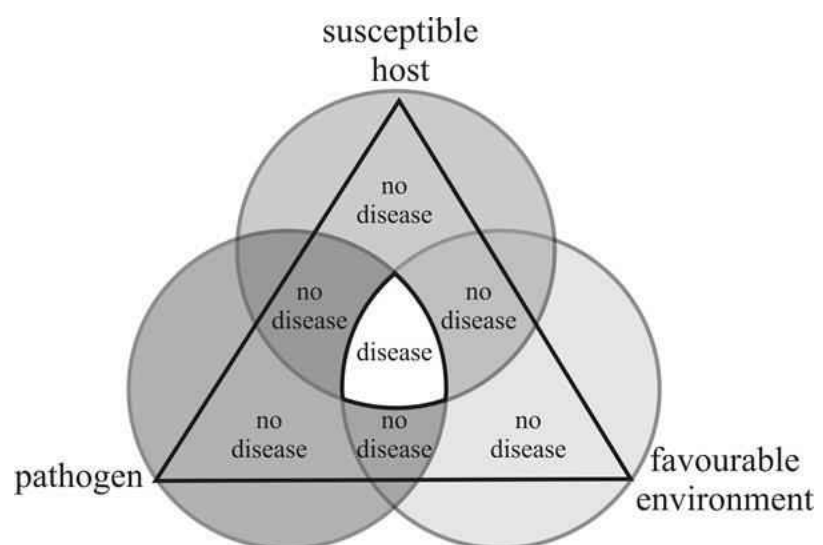
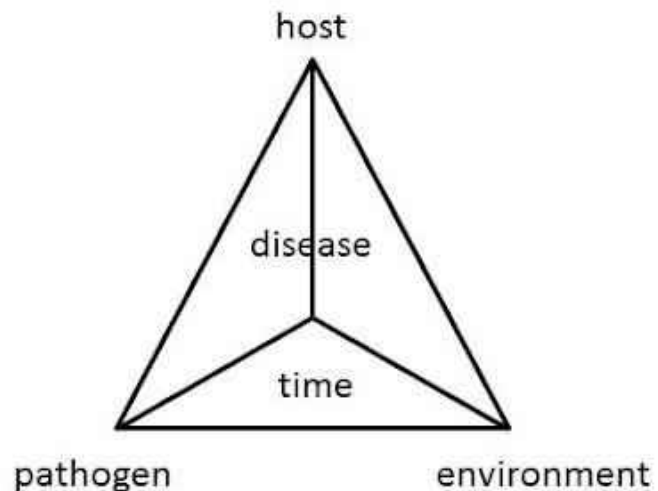


illustration of the continuum of host reaction from complete susceptibility to immunity, and the degree of pathogen virulence, and the environmental suitability for disease. If any one element is reduced to zero the triangle transforms into a line and the area occupied by disease collapses to zero. Less dramatic alterations in any factor change the area of the central disease envelope which is an indicator of disease intensity (incidence or severity). For example, a host with some degree of resistance will have a smaller susceptibility circle, and consequently lesser area of overlap and less severe disease. Another example could be a pathogen with greatly increased virulence, which would be shown as a larger 'pathogen circle' and consequently larger area of overlap and more severe disease.

The four-dimensional disease pyramid with “time” added.

4. Time

Disease does not occur instantaneously, even in situations where a conducive environment remains present for an extended period of time. Time is required for the pathogen to infect the plant and produce characteristic lesions as well as produce sporulation (the asexual phase of reproduction). A prolonged period of time can also elapse between infection of the plant and symptom expression. Specific diseases where this likely tends to occur include *Cercospora* leaf blight and soybean rust.



In addition, in certain disease situations an extended period of time may be required for infection, disease development, and continued sporulation to occur. As I said above, disease incidence is not instantaneous. Several days may be required to go through several stages of reproduction.

Time is an essential dimension and has been added to the disease triangle by several authors, primarily to convey the idea that disease onset and intensity are affected by the duration that the three prime factors are aligned. Some duration of favourable alignment is necessary for disease to occur; but the length of time depends on your level of analysis. Physiological events in the host that define infection can take place in minutes or hours; disease symptoms in the field can take days or weeks to appear. Showing time as a dimension on the triangle (perhaps converting it into a pyramid) could be a more realistic adaptation of the diagram.

Boom and Bust Cycle and Vertifolia Effect

Boom and Bust Cycle:

It may be expected that when a new variety carrying VR is developed, it is resistant to the prevalent Pathotypes of the pathogen in question. Therefore, it becomes popular rapidly and occupies a large proportion of the growing area under the concerned crop. This period of boom (high popularity and cultivation in a large proportion of area) continues till the Pathotypes specific to the VR gene (s) employed in the variety evolves and /or increases in frequency to cause an epidemic in the field occupied by the variety.

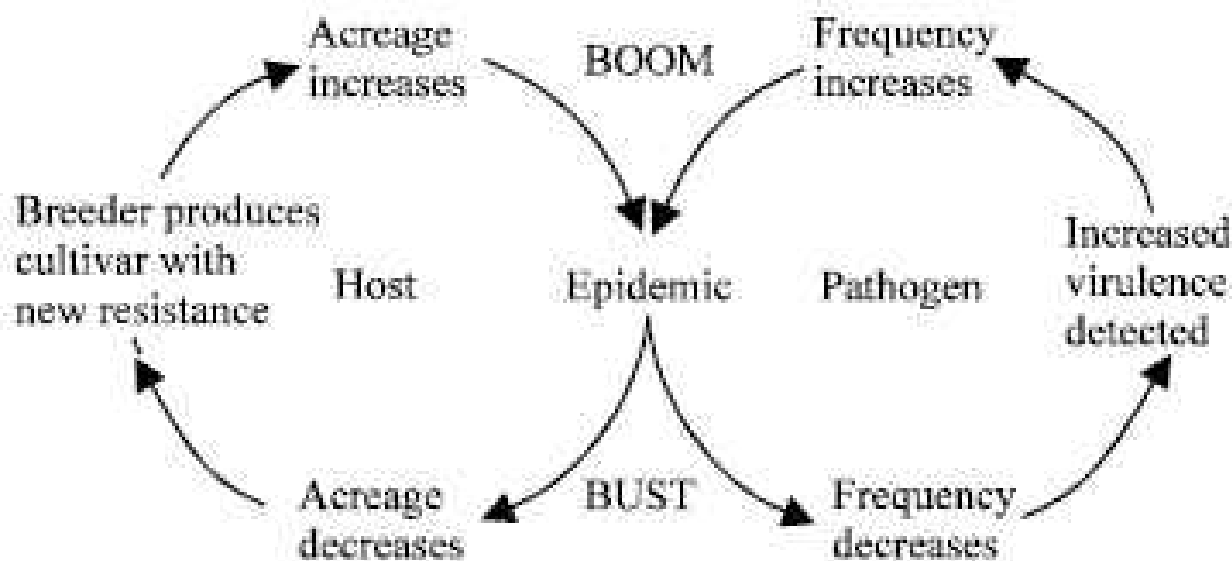
The now susceptible variety loses popularity and area under it declines sharply thus the variety is effectively busted by the virulent Pathotypes. This has been called 'boom and bust' cycle by Priestey (1970). When a variety goes bust, breeders develop new variety with a new VR gene, which makes it resistant to the predominant Pathotypes.

This variety also undergoes the 'boom and bust' cycle, and the breeders put forth a new variety and so on. The period for which such a variety may last is, on an average, -5 years in the case of air-borne fungal pathogens. But in many cases, this period is much shorter, in extreme cases, the variety go bust even before it reaches farmers' fields.

The reason for boom and bust cycle line in the genetic system of the pathogen and the selection pressure exerted by the host on the pathogen Pathotypes. When a resistant variety becomes popular it is cultivated on large areas. The plants of this variety act as strong selection agents for the Pathotypes specific to the new variety since only this Pathotypes can cause disease in the plants of this variety. Therefore, the specific Pathotypes increases rapidly in frequency in the pathogen population.

This specific Pathotypes may be newly evolved or may have already been present in the pathogen population, albeit, in a very low frequency. Soon the frequency of the virulent Pathotypes reaches a stage at which it causes an epidemic in the thus far resistance variety. Usually, the epidemic would occur after a couple of or more years following the first detection of the virulent Pathotypes.

The popularity of and the area under the variety will decline rapidly following the epidemic. This will reduce or eliminate the selection pressure in favour of the Pathotypes virulent on the variety. As a result, the frequency of this Pathotypes in pathogen population will also decline.



Vertifolia Effect:

Van der Plank introduced the term Vertifolia effect derived from the German potato variety 'Vertifolia' having the late blight resistance genes R3 and R4. The variety became susceptible when the appropriate Pathotypes P(3,4) evolved in the pathogen leading to a nearly complete failure of the crop; such a total failure of vertical resistance leading to a disease epidemic is known as Vertifolia effect. Thus Vertifolia effect refers to an epidemic development in a variety carrying vertical resistance genes and a low level of HR, leading to heavy economic losses. The failure occurs because of the following two reasons: 1) The level of horizontal resistance in the varieties carrying oligogenes for resistance is usually low, and 2) The pathogen is able to evolve the virulent Pathotypes specific to the oligogenes for vertical resistance.

Lec. 3. Disease surveillance – methods of survey- surveillance report- diseases surveillance programs in Tamil Nadu.

Plant Disease Survey

Disease survey is one of the main aspects in the disease management studies. The plants disease survey is very useful (1) to know the prevalence of disease in a crop in a particular area, (ii) to study the damages caused by a particular disease and assess the economic importance of the disease, (iii) to correlate the disease incidence with the weather factors, vegetation and soil factors, (iv) to give suitable forecasting to the farmers about the outbreak of file disease and (v) to device suitable control measure in advance by cultural and chemical means.

In Tamil Nadu, a well organised Pest Surveillance Programme is functioning throughout the state. The officials of the Department of Agriculture and Department of Horticulture and Plantation crops are doing the survey in collaboration with the scientists of Tamil Nadu Agricultural University and they have sent weekly pest/disease surveillance report to the University Analysis and compilation. The pest/disease surveillance report should also accompany the details on the daily weather factors, viz., maximum temperature, minimum temperature, relative humidity, wind velocity, rainfall, rainy days, etc.

METHODS OF SURVEY

(i) Fixed plot survey

Select two villages and fix two fields per village. Select five sampling plots per field of one acre and it should be three meters away from the bunds. The size of the sample plot should be 1 Sq. m. Select 20 plants or leaves per plot and observe for the disease intensity using the standard disease score chart having 0-9 grades. Work Out the percent disease index (PDI) using the standard formula. Survey the field every week and prepare the report.

(ii) Roving survey

Select four villages and in each village, fix two fields. Observe and score 1 00 plants or leaves in each field by walking across, starting from south west corner to North East corner. Work out the PDI using the standard formula. Survey the fields every week and prepare the report.

DISEASE SURVEILLANCE REPORTS

(i) White Report

It contains the disease/pest surveillance detail in a particular block/division/district for a week.

(ii) Yellow Card

It should be sent when the disease/pest occurrence attained half the level of Economic Threshold Level (ETL).

(iii) Red Card

It should be sent when the disease/pest occurrence exceeded the Economic Threshold Level.

The weekly data on disease incidence along with the weather parameters collected from fixed plot and roving surveys are reported to the Joint Director of Agriculture in each district. These data collected from the farmers' fields and research centres in the district are pooled and interpreted jointly by the plant protection scientists of the university and Deputy Director of Agriculture (Plant Protection) every week and the action to be taken is decided. After analysing the weekly survey reports and the data on the weather factors, the message on the severity level of disease and control measures to be adopted are

disseminated to the farmers through mass media. So surveillance and monitoring methods must be continuously adopted to trace the diseases, their movement and spread with a view to prevent the epidemics.

Lec 4. Assessment of Plant Diseases- methods of Assessment- measurement of diseases growth rate by Area Under Disease Progressive Curve (AUDPC)

Plant Disease Assessment

The assessment of the amount of disease on a plant, more importantly on a crop, is essential in any quantitative epidemiological study. Disease assessment also forms the fundamental basis of many other aspects of plant pathology. Assessment data are essential to breeders, fungicide manufacturers, economists, government agencies and academics in their various evaluations of resistance, treatment efficacy and resource priorities.

With such diverse investigational objectives, the precision of the assessment methods and the quality of data collected will vary accordingly. Inevitable, as large areas may be involved, sampling will be carried out, a compromise being necessary between what ideally should be done and what it is practical and economical to do.

Most importantly, assessment of a particular disease in an individual crop over several years can provide indicators to elucidate the factors governing its incidence and severity. This information can also used to devise forecasting systems.

Methods of Assessment

1. Assessment in terms of percentage

This method is applicable for those diseases which complete death of plants (eg. damping off, root rot, wilt, etc.). This also followed in systemic diseases (eg. Virus and MLO diseases, etc.) and in the diseases leading to total destruction of infected organs (eg. Smut, green ear, etc.) The per cent disease incidence is calculated using the following formula,

$$\text{Per cent disease Incidence} = \frac{\text{Number of infected plants}}{\text{Total number of plants observed}} \times 100$$

2. Assessment using disease grades

This method is useful to assess the foliar diseases like downy mildew, powdery mildew, leaf spot, leaf blight, canker, rust, etc. A of disease with area affected or per cent grains affected, indicating the grades anging from 0 to 9. The Plants are observed individually and severity percentages are arrived and the numerical rating is given as 0, 1, 3, 5, 7 or 9 as shown below. Normally, 50 or 100 plants or leaves are observed and individual rating is given.

Eg. Paddy - Blast (*Pyricularia oryzae*)

Grade

- 0 No visible symptoms
- 1 < 1% leaf area affected
- 3 1-10% leaf area affected
- 5 11-25% leaf area affected
- 7 26-50% leaf area affected
- 9 >50% leaf area affected

Eg. Sorghum-rust (*puccinia purpurea*)

Grade

- 0 No visible symptoms
- 1 < 1% leaf area affected with pustules growth
- 3 1-10% leaf area affected with pustules
- 5 11-25% leaf area affected with pustules
- 7 26-50% leaf area affected with pustules
- 9 >50% leaf area affected with pustules

Eg. Cumbu-Ergot (*Claviceps fusiformis*)

Grade

- 0 Earheads free from infection
- 1 < 1% grains in earhead replaced by sclerotia
- 2 1-10% grains in earhead replaced by sclerotia
- 5 11-25% grains in earhead replaced by sclerotia
- 7 26-50% grains in earhead replaced by sclerotia
- 9 >50% grains in earhead replaced by sclerotia

Eg. Grapes-Downy mildew (*Plasmopara viticola*)

Grade

- 0 No visible symptoms
- 1 <1% leaf area with mildew growth
- 2 1-10% leaf area with mildew growth
- 5 11-25% leaf area with mildew growth
- 7 26-50% leaf area with mildew growth
- 9 >50% leaf area with mildew growth and severe infection on fruits

The sampling unit and sample size will vary according to the type of experiment, nature of host and disease involved. In the case of foliar diseases where the assessment of all the plants in a plot or a field is not possible, 1 per cent representative population is selected at random for assessment. In the case of tree crops, five trees are selected at random and 20 representative leaf or fruit sample are assessed at random. In each tree using the standard disease score chart. The percent disease index is worked out using the following formula.

$$PDI = \frac{\text{Sum of individual ratings}}{\text{Total number of plantsgrade/leaves observed}} \times \frac{100}{\text{maximum disease}}$$

The assessment of the disease intensity and evaluating yield loss are important factors with regard to a plant disease in a particular crop. This is important to growers to warrant the trouble due to a disease and to take up control measures. Therefore, knowledge about nature of disease, its intensity and economic importance is necessary to forecast a disease.

Area Under the Disease Progress Curve (AUDPC)

Calculating the area under the disease progress curve to quantify disease progress

The area under the disease progress curve (AUDPC) is a useful quantitative summary of disease intensity over time, for comparison across years, locations, or management tactics. The most commonly used method for estimating the AUDPC, the trapezoidal method, is to discretize the time variable (hours, days, weeks, months, or years) and calculate the average disease intensity between each pair of adjacent time points (Madden *et al.* 2007). We can consider the sample time points in a sequence $\{t_i\}$, where the time interval between two time points may be consistent or may vary, and we also have associated measures of the disease level $\{y_i\}$. We define $y(0) = y_0$ as the initial infection or the disease level at $t = 0$ (i.e., the first disease severity observation in our study). $A(t_k)$, the AUDPC at $t = t_k$, is the total accumulated disease until $t = t_k$, given by . To illustrate this concept, the exercise below shows the rectangular area S_i as the increase in accumulated disease during the time interval from t_{i-1} to t_i , $S_1 = A_1$ and $A_i = A_{i-1} + S_i$, $i \geq 2$.

For the purposes of this exercise we will write our own AUDPC function. The [Agricolae package](#) for R also includes an AUDPC function. Below is an illustration of the function to calculate the AUDPC and graph the output.

```
#Set up disease severity percent measurements;
#change these in subsequent analyses to see how it affects
#the AUDPC
ds0<-1
ds1<-2
ds2<-7
ds3<-7.5

#Put these values into a vector without making any changes
disease.severity<-c(ds0,ds1,ds2,ds3)

#Time points at which disease severity
# measurements are made,
#change these in subsequent analyses to
#see how it affects the AUDPC Value
t0<-0
t1<-2
t2<-5
t3<-6

#Put time period into a vector
## Do not change these values
```

```
time.period<-c(t0,t1,t2,t3)
```

#Refresh your memory about how the plot function works
Help (plot)

#Create the plot of disease severity over time

```
Plot (time.period,  
      disease.severity,  
      ylim=c(0,(ds3+1)),  
      xlim=c(0,(t3+0.5)),  
      xlab="Time",  
      ylab="Disease Severity (%)",  
      type="o",  
      pch=19,  
      col="mediumblue")
```

#Add a title and subtitle to our plot

```
title(main="Illustration of AUDPC Calculation",  
      sub="Figure 1")
```

#Add text to x labels defining time periods

defined in text

```
mtext("=t0",1,at=0.3,1)  
mtext("=t1",1,at=2.3,1)  
mtext("=t2",1,at=5.3,1)  
mtext("=t3",1,at=6.3,1)
```

#Illustrate the area under disease progress

curve with rectangles.

Do not change these values

```
rect(t0,0,t1,((ds0+ds1)/2),border="orange")
```

Add text to rectangle to describe rectangle

```
text(1,1,"A1")
```

#Add segment to Y axis

#And so-on

```
rect(t1,0,t2,((ds1+ds2)/2),border="orange")
```

```
text((t1+t2)/2,(((ds1+ds2)/2)/2),"S2")
```

#Draw line to axis and label with value

```
segments(.4,((ds1+ds2)/2),t2,((ds1+ds2)/2),  
          col="black",lty="18")
```

```
text(0,((ds1+ds2)/2),((ds1+ds2)/2))
```

```
rect(t2,0,t3,((ds2+ds3)/2),border="orange")
```

```
text((t2+t3)/2,(((ds2+ds3)/2)/2),"S3")
```

```
segments(0.4,((ds2+ds3)/2),t2,((ds2+ds3)/2),  
          col="black",lty="18")
```

```
text(0,((ds2+ds3)/2),((ds2+ds3)/2))
```

#Build a function for AUDPC calculation

#the left curly bracket indicates the beginning

of the function

```

audpc <- function(disease.severity,time.period){

  #n is the length of time.period, or
  # the total number of sample dates
  n <- length(time.period)

  #meanvec is the vector (matrix with one dimension)
  #that will contain the mean percent infection
  #it is initialized containing -1 for all entries
  #this sort of initialization is sometimes useful
  # for debugging
  meanvec <- matrix(-1,(n-1))

  #intvec is the vector that will contain the length of
  # time between sampling dates
  intvec <- matrix(-1,(n-1))

  #the loop goes from the first to the penultimate entry
  #the left curly bracket indicates the beginning of
  # commands in the loop
  for(i in 1:(n-1)){

    #the ith entry in meanvec is replaced with the
    # mean percent infection
    #between sample time i and sample time i+1
    meanvec[i] <- mean(c(disease.severity[i],
                        disease.severity[i+1]))

    #the ith entry in intvec is replaced with the length
    # of the time interval between time i and time i+1
    intvec[i] <- time.period[i+1] - time.period[i]

    #the right curly bracket ends the loop
  }

  #the two vectors are multiplied together
  # one entry at a time
  infprod <- meanvec * intvec

  #the sum of the entries in the resulting vector
  # gives the AUDPC
  sum(infprod)

  #the right curly bracket ends the function
}

#Now apply the function to the example data and put
# the result in a new object called 'AUDPCexample'
audpc(disease.severity,time.period) -> AUDPCexample

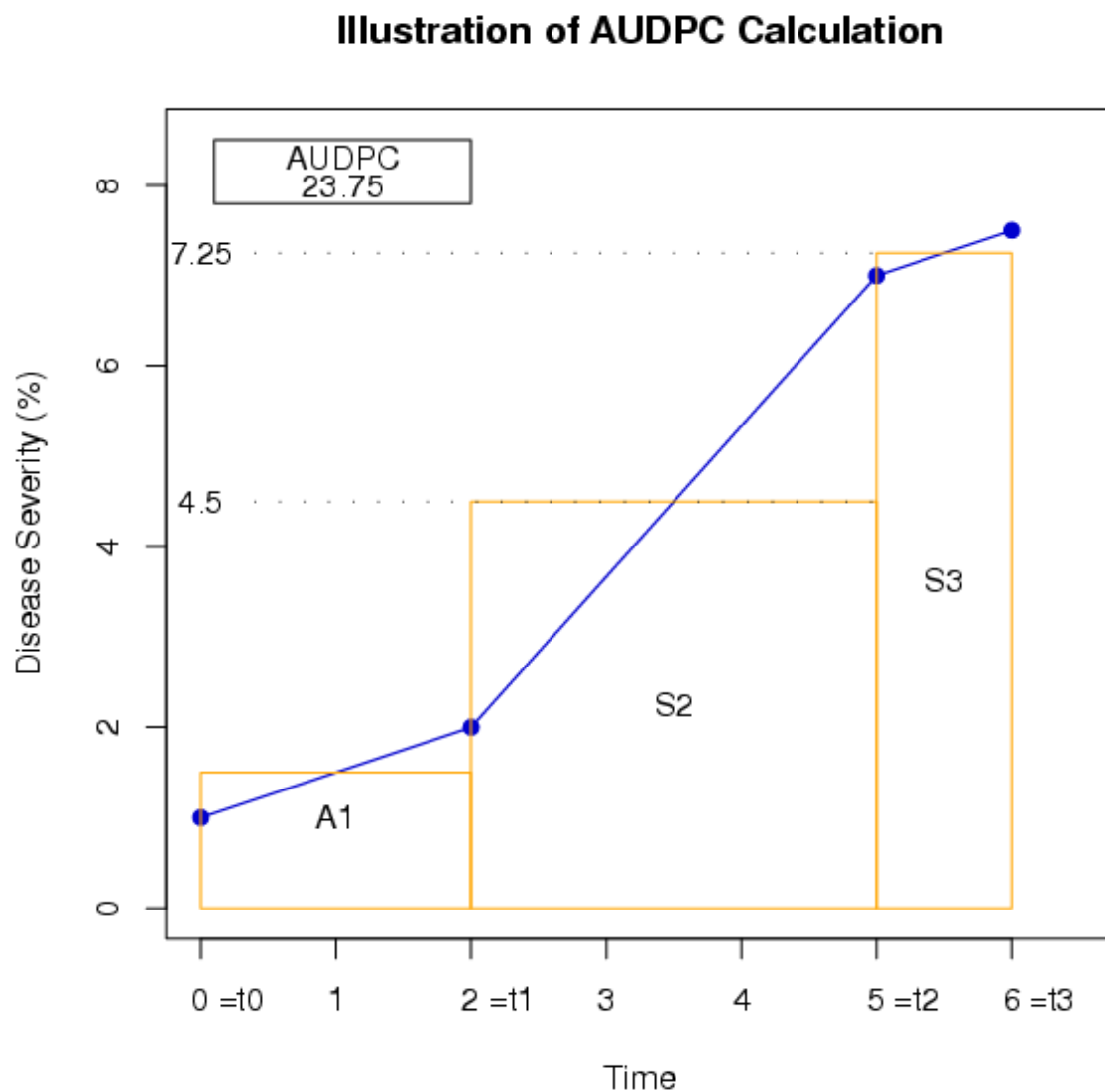
```

```

#Display AUDPC Value
#Draw rectangle around value
rect(0.1,(ds3+.3),2,(ds3+1),border="black")
#AUDPC Text
text(1.05,(ds3+0.8),"AUDPC")
text(1.05,(ds3+0.5),AUDPCexample)

```

Output



You can use the R code above to illustrate other examples by replacing the values of ds0 through ds4 and t0 through t4, as commented in the R script above. The function for calculating the AUDPC, can be used with other data sets for which the number of disease observations is equal to the number of time points and for which the time points appear in order. Since you might have your own disease progress data, this would be a good time to revisit how to use the read.csv function. Refer to [An Introduction to the R Programming](#)

[Environment](#) for a brief tutorial on how to import data from spreadsheets and other sources (Garrett *et al.* 2007).

Lec 5. Forecasting of Plant Diseases- Forecasting Models in Plant Diseases Management

Forecasting of plant diseases means predicting for the occurrence of plant disease in a specified area ahead of time, so that suitable control measures can be undertaken in advance to avoid losses. Disease forecasts are predictions of probable outbreaks or increase in intensity of disease. It involves well organized team work and expenditure of time, energy and money. It is used as an aid to the timely application of chemicals. Among the first spray warning services to be established for growers, were the grapevine downy mildew forecasting schemes in France, Germany and Italy in the 1920s. Disease forecasting methods are available for the following plant diseases.

Sl. No	Plant disease	Countries
1.	Grapevine downy mildew	Australia, France, Germany, Greece, Italy, Romania, Spain, USSR, Yugoslavia
2.	Cucurbit downy mildew	U.S.A.
3.	Potato late blight	Australia, Brazil, Finland, France, Germany, Greece, Japan, the Netherlands, Norway, Peru, U.K, the U.S.S.R.
4.	Tobacco blue mould	Canada, U.S.A.
5.	Apple and pear scab	Australia, Canada, Netherlands, New Zealand, the U.S.A.
6.	Sugarbeet root rot (<i>Aphanomyces</i> sp.)	U.S.A
7.	Wheat brown (Leaf) rust	U.S.A.
8.	Corn bacterial wilt (<i>Erwinia stewartii</i>)	U.S.A
9.	Sugarbeet curly top	U.S.A.

Informations needed for disease forecasting

Forecasting diseases is a part of applied epidemiology. Hence, knowledge of epidemiology (development of disease under the influence of factors associated with the host,

pathogen) is necessary for accurate forecasting. The factors of epidemic and its components should be known in advance before forecasting is done. The informations required for forecasting are:

1. Host Factors

- a. Prevalence of susceptible varieties in the given locality
- b. Response of host at different stages of the growth to the activity of pathogen e.g. Some diseases are found during seedling stages while others attack grown up plants and
- c. Density and distribution of the host in a given locality. Dense populations of susceptible variety invite quick spread of an epidemic. Growing susceptible varieties in scattered locations and that too in a limited area are less prone to epiphytotic.

2. Pathogen factors

- a. Amount of primary (initial) inoculum in the air, soil or planting material
- b. Dispersal of inoculum
- c. Spore germination
- d. Infection
- e. Incubation period
- f. Sporulation on the infected host
- g. Re-dispersal / Dissemination of spores
- h. Perennating stages
- i. Inoculum potential and density in the seed, soil and air

3. Environmental factors

- a. Temperature
- b. Humidity
- c. Light intensity
- d. Wind velocity

Requirements or conditions for disease forecasting

There are five main requirements which must be satisfied before a useful and successful disease forecast is made.

1. The disease must cause economically significant damage in terms of yield loss or quality. Damage assessment is essential to develop strategy for controlling a disease. e.g., Annual estimation of yield loss caused by barley powdery mildew (*Erysiphe polygoni*) in England and Wales had ranged from 6 to 13 %. Potato late blight can cause a yield loss of 28% if the disease reaches the 75% stage by mid-August. Diseases like apple scab and potato common scab reduces the quality of the produce lower the value of the harvested crop and cause considerable financial loss to the growers.
2. Control measures must be available at an economically acceptable cost.
3. The disease must vary each season in the timing of the first infections and its subsequent rate of progress. If it does not, there is no need for forecasting.

4. The criteria or model used in making a prediction must be based on sound investigational work carried out in the laboratory and in the field and tested over a number of years to establish its accuracy and applicability in all the locations where its use is envisaged.
5. Growers must have sufficient man power and equipments to apply control measures when disease warning is given. Long-term warnings or predictions are more useful than short-term warning or predictions.

Methods of disease forecasting

Disease forecasting requires field observations on the pathogen characters, collection of weather data, variety of the crop and certain investigations and their correlations. Usually the following methods are employed in disease forecasting.

1. **Forecasting based on primary inoculum:** Presence of primary inoculum, its density and viability are determined in the air, soil or planting material. Occurrence of viable spores or propagules in the air can be assessed by using different air trapping devices (spore traps). In the case of soil-borne diseases the primary inoculum in the soil can be determined by monoculture method. Presence of loose smut of wheat, ergot of pearl millet and viral diseases of potato can be detected in the seed lots at random by different seed testing methods. Seed testing methods can be used to determine potential disease incidence and enable decision to be made on the need for chemical seed treatment. The extent of many virus diseases is dependent on the severity of the preceding winter which affects the size of vector population in the growing season. e.g., Sugarbeet yellows virus.
2. **Forecasting based on weather conditions:** Weather conditions viz., temperature, relative humidity, rainfall, light, wind velocity etc., during the crop season and during the inter crop season are measured. Weather conditions above the crop and at the soil surface are also recorded.
3. **Forecasting based on correlative information:** Weather data of several years are collected and correlated with the intensity of the diseases. The data are compared and then the forecasting of the disease is done. Forecasting criteria developed from comparisons of disease observation with standard meteorological data have been provided for diseases like *Septoria* leaf blotch of wheat, fire blight of apple and barley powdery mildew.
4. **Use of computer for disease forecasting:** In some advanced countries forecasting of disease is made by the use of computers. This system gives the results quickly. One such computer based programmes in the USA is known as '**Blitecast**' for potato late blight.

Examples of well-developed forecasting systems are given below.

- a. **Early and late leaf spots of groundnut:** A technique has been developed for forecasting early and late leaf spots of groundnut in the U.S.A. When the groundnut foliage

remains wet for a period greater than or equal to 10 h and the minimum temperature is 21°C or higher for two consecutive days or nights, the disease development is forecasted.

A computer programme has been developed in the USA. This is accurate and is widely used in the USA. The data on hours for day with relative humidity (RH) of 95% and above and minimum temperature (T) during the RH observations for the period, for the previous 5 days are fed to the computer. Calculations are rounded to whole numbers. The T/RH index for each of the five days is calculated e.g., when hours of the RH 95% equal 10 and the minimum temperature during the period equals 21.1°C the T/RH index is 2.0. The T/RH indices for days 4 and 5 are summed. If the total index exceeds 4 disease is forecasted. If the index is 3 or less no disease is forecasted.

b. Late blight of potato: In the USA a forecasting programme has been developed for late blight of potato (*Phytophthora infestans*). The initial appearance of late blight is forecasted 7 to 14 days after the occurrence of 10 consecutive blight favourable days. A day is considered to be blight favourable when the 5 days average temperature is 25.5°C and the total rainfall for the last 10 days is more than 3.0 cm.

A computerized version (**Blitecast**) has also been developed in the U.S.A for forecasting potato late blight. Blitecast is written in Fortran IV. When a farmer desires blight cast (blitecast) he telephones the blight cast operator and reports the most recently recorded environmental data. The operator calls for the blight cast programmes in the computer viz., typewriter terminal and feeds the new data into the computer. Within a fraction of second the computer analyses the data and series of a forecast and spray recommendations to the operator who relays it to the farmer. The entire operation can be completed during standard three minutes telephone call. The system makes one of the four recommendations viz., no spray, late blight warning, 7 days spray schedule or 5 days spray schedule. The last 5 days spray schedule is issued only during severe blight weather.

In West Germany, '**Phytoprog**' is the programme used. It is based on measurements of temperature, relative humidity and rainfall. Phytoprog provides a negative prognose (an indication of when the usual routine spray application should be dispensed with).

c. Blister blight of tea: A system for predicting epidemics of blister blight of tea (*Exobasidium vexans*) has been developed based on the number of spores in the air in the tea plantation and the duration of surface wetness on the leaves. The duration of sunshine is negatively correlated with the duration of surface wetness. The following prediction equation has been developed.

$$Y = 1.8324 + 0.8439 X_1 + 0.9665 X_2 - 0.1031 X_3$$

where, X_1 = log % infection t_2

X_2 = log % infection t_2 – log infection t_1

Y = log of the number of spores in the air and $t_1 - t_2$ three weeks

X_3 = mean daily sunshine for a 7 days period preceding t_2

d. Southern corn leaf blight: '**Epimay**' is a system for forecasting Southern corn leaf blight (*Bipolaris maydis*) based on conceptual model.

e. Rice blast: In India, forecasting rice blast (*Pyricularia oryzae*) is done by correlative information method. It is predicted on the basis of minimum night temperature 20 to 26°C in association with high relative humidity of 90% or above. Computer based forecasting system has also been developed for rice blast in India.

f. Wheat stem rust: Forecasting wheat stem rust epidemic is done by analysing the rain samples which give precise data for inoculum present in the air. Moreover several wind trajectories are also prepared to survey the air-borne primary inoculum and its deposition. It has been observed that primary inoculum comes from South India, to the plains of Central and North India.

g. Brown stripe downy mildew of corn: The forecasting of brown stripe downy mildew of corn (*Sclerophthora raysise* var. *zeae*) which is restricted to India is done on the basis of average rainfall 100 to 200 cm or more accompanied by low temperature (25°C or less).

h. Apple scab forecasting in Himachal Pradesh

- In Himachal Pradesh, when the apple scab appeared in epidemic form in late 1970's onward, 5 Apple Scab Monitoring and Research Laboratories were established at Mashobra, Kotkhai, Thanedhar and Sarahan in Shimla, and Bajaura in Kullu district.
- Apple scab infection periods were predicted by using equipments, like de Wit 7 day leaf wetness recorder, Reuter Stokes apple scab predictor and SWG Biomat at different locations.
- When the conditions congenial for the development of apple scab were recorded, this information was given to the farmers by blowing sirens, radio broadcasts and personal contacts.
- Of all the systems, Reuter Stokes predictor was found to be the most efficient in predicting the infection periods at Kotkhai (Sharma and Gupta, 1995).
- Different fungicides could be effectively used depending on their after-infection (curative), pre-symptom or post-symptom (eradivative) activities. This is regarded as the best example of plant disease forecasting in India till date.

Lec. 6. Diagnosis of plant diseases-seed health test, chemo diagnosis, Serodiagnosis, and molecular diagnosis.

1. Seed health test

Seeds carry several destructive pathogens that often take a heavy toll by causing severe diseases on crops raised from them. Disease agents may be carried externally or internally. Disease agents carried with seeds may be fungi (e.g. loose smut of wheat, brown spot of rice) or bacteria (e.g. black leg of cabbage, black arm of cotton) or virus (e.g. bean common mosaic) or even nematodes (e.g. ear cockle of wheat, paddy white tip). So seed health testing is very essential to obtain the information on health

status of the seed lot. Methods of testing seeds for the presence of pathogens are of four main types. The seeds may be (i) directly examined, (ii) examined after incubation, (iii) plated on agar so that the pathogen grow out into colonies and (iv) subject to techniques which allow the use of modern serological developments viz., ELISA and others.

Seed Health Testing Methods

1. Examination of dry seeds

It is applied for detection of seed borne fungal pathogens which cause discoloration of the seed or change the shape and size of the seed. Also applicable for detecting fungal structures present in, on or with seed. This method gives the information on physical purity of the seed lot. It is suitable to detect the sclerotia, galls, smut balls, discolouration, malformation, resting hypha, fruiting bodies of fungi (oospores, smut spores, pycnidia, perithecia) and bacterial masses. A sample of 400 seed (in replication of 100 seeds each) may be drawn and examined for the presence of discoloured, malformed and diseases affected seeds. The number is recorded and the percentage is worked out on the number basis.

Example : Ergot of bajra (*Claviceps fusiformis*), Karnal bunt of wheat (*Neovossia indica*).

2. Seed washing technique

This is useful in testing surface-borne, contaminating fungi like smuts, bunts, downy mildews, powdery mildews, rusts, etc. Two gram seed sample is taken in a test tube and mixed well by adding 2 ml of sterile water, for 5 to 10 minutes. The supernatant is taken and centrifuged at 2000 rpm for 10 minutes. The supernatant solution is observed for the fungal structures under a microscope.

Example : Brown spot of rice (*Bipolaris oryzae*), Flag smut of wheat (*Urocystis agropyri*)

3. Blotter method

This method is widely used. All kinds of cereals, vegetables, crucifers, legumes, ornamentals and forest seeds are tested by this method.

In this method, the seeds are placed on moistened blotters (three layers of filter paper in a petridish form a blotter). Generally 400 seeds are drawn for this test, with replication of 100 seeds each. Twenty five seeds are placed in blotters at equal distance. The blotters are incubated for 7 days under alternating cycle of Near Ultra Violet (NUV) light (12 hr.) and darkness (12 hr.) at 20 °C. Then the fungal growth developed on the seeds are examined and identified under stereoscopic microscope.

Freezing blotter method

It is a modified blotter method, wherein the blotters are incubated at 20°C for 2 days, then at - 20°C for a day and again at 20 °C for 5-7 days.

2,4-D Blotter method

In this method, the seeds are placed in blotters added with 0.1 to 0.2% 2,4-D solution and incubated as above.

4. Agar plate method

In this method, the seeds are treated with 1% sodium hypochlorite solution (to prevent the growth of saprophytic fungi for 5-10 minutes. Then, the seeds are placed on potato dextrose agar medium or malt extract agar medium and incubated at $20 \pm 2^{\circ}\text{C}$ for 7 days. The plates are examined periodically for the fungal growth to identify the pathogen.

5. Grow on test /Seedling symptom test

This test is applicable for those fungi which are capable of producing symptoms on the root and shoot of the young seedlings. This test for certain pathogens, provide information pertaining to field performance of the seed lot. Seeds are sown in autoclaved soil or sand or any type of other media and incubated at 20°C for 14 days under 12h of alternating cycles of artificial light and darkness After incubation, individual seedling is examined and per cent infection is calculated. Grow on test is normally used to determine virus seed transmission.

6. Indicator test

Viruses can be detected by assaying the extracts of different parts of seeds and seedlings raised from infected seeds on suitable indicator plants.

2. Chemo diagnosis

It is a technique of identification of plant pathogens (viruses) using certain chemicals. Chemical methods to identify certain plant viral diseases have been well established.

i. Tetrazolium chloride test

This method is specially followed for detection of *Banana bunchy top virus* and cassava mosaic viruses. Thin sections are taken from the mid rib or petiole of the leaf and soaked in 5 ml of 1% solution of 2, 3, 5- triphenyl tetrazolium for 30-60 minutes in a test tube. The sections

are then removed and observed for coloration. Sections turn to brick red colour, indicates the infection of virus in the plant sample. If the sections remain green, indicates the healthy nature of the plants. This helps in selection of disease free planting materials.

ii. Iodine test

This test is used to identify rice tungro virus infection. Iodine solution is prepared by dissolving 2 g of Iodine and 6 g of potassium iodide in 100 ml of water. The iodine solution is taken in a test tube and 10 cm long leaf tip is dipped in the solution for 15 minutes (collect the leaf samples before 6 am in the morning). Then, the leaf tip is taken out and washed thoroughly with water and examined. Tungro virus affected leaves show blue streaks, while the healthy leaves retain their green colour. A solution containing 10 ml of tincture iodine in 140 ml of water can also be used for this test. In this case, leaf samples should be dipped for 1 hour.

iii. Paraquat test

The herbicide paraquat (1,1'-dimethyl4,4'-bipyridinium dichloride) has been used to enhance the detection of latent infection of species of *Colletotrichum*, *Alternaria*, *Botrytis*, *Cercospora*, *Phomopsis* and other fungi on healthy appearing plant material. Chlorophyll-containing tissues are killed rapidly by paraquat, a process that accelerates the growth and sporulation of fungal colonists and some endophytes. Plant parts are treated with 0.03 to 0.6% of the paraquat solution and incubated under humid conditions to encourage the growth and sporulation of fungal colonists, which are subsequently identified.

iV. Diagnosis based on Ooze Test

Ooze test is used to quickly diagnose bacterial diseases. Collect the disease infected leaves and cut a young lesion across and place in a transparent glass slide containing a drop of clear water. After a few minutes, hold the slide against light and observe for thick or turbid liquid coming from the cut end of the leaf.

Collect the disease infected stem and surface sterilize with 10% Sodium hypochlorite. Cut the infected stem and dipped into the glass beaker containing clear water. Bacterial ooze will comes out from the cut end and make the water to become turbid.

3. Serodiagnosis (Serological tests)

The serological tests can be performed by *combining antigen (virus)* and its *antibody (antiserum)* in several ways for the detection and identification of antigenic substances and the organism that carry them.

Protein Based Techniques – These techniques detect the virus coat protein by agglutination reaction or antigen-antibody reaction. Based on the phase in which these reactions occur, these are further classified as –

Liquid Phase Tests – In these tests, the reactions occur in a semi-solid medium such as Agar. These tests include-

- Precipitin Tests
- Agglutination Tests
- Ouchterlony Double Immunodiffusion Test

Solid Phase tests – In these tests, the reactions occur in a solid medium such as polystyrene plate or on a membrane such as Nitrocellulose membrane or Nylon Membrane.

- Enzyme Linked Immunosorbent Assay (ELISA)
- Sodium Dodecyl Sulphate – Poly Acrylamide Gel Electrophoresis (SDS-PAGE)
- Immunosorbent Electron Microscopy (ISEM)
- Western Blot
- Dot Immuno Binding Assay (DIBA)
- Tissue Immuno Binding Assay (TIBA)

1. Precipitin tests : When antigen and antibody are mixed they combine and form a precipitate. This precipitation or precipitin reaction is widely used in Plant Virology. The extent of precipitate formed is dependent on a number of factors, e.g., salt concentrations, pH, temperature and presence of interfering compounds and the ratio of concentration of antibody and antigen is important as, excess of either may prevent precipitation. Most rapid precipitation occurs at a definite antigen/antibody ratio called the *optimal proportion ratio*. The greatest dilution of antigen (virus) to produce visible precipitate is called *antigen dilution-end point*. The greatest antiserum (antibody) dilution giving a visible precipitate is the *antiserum titre*. There are number of precipitin tests. The important ones are as follows:

A. Precipitin tests in liquid media

- (a) Tube precipitation test :** Various dilutions of clarified virus suspension and antiserum (1:10, 1:20, 1:40, 1:80, 1:160) are mixed in narrow thin walled glass tubes and incubated at 37°C in a water bath. The resulting precipitate (min. /hrs.) is flocculent type with elongated particles and densely granular with spherical particles. Relatively higher concentration of reagents is needed in the precipitation reaction.
- (b) Tube (ring interface) precipitation :** It is modification of the precipitation tube test. Virus suspension (0.25 ml to 0.5 ml) is mixed with an equal volume of antiserum in small test tubes (0.7 cm dia.). Various dilutions (1:10, 1:20, 1:40, 1:80, and 1:160) of each reactant (i.e. antigen and antiserum) can be tested against each other. Usually antiserum is placed in a small tube and antigen is then carefully layered on top. The sample can be incubated at 37°C in tubes only, partially immersed in the water bath in order to cause mixing by convection. Some plant protein (in a partially purified virus) precipitates at this temperature. Pre-treatment of sap, if sap is to be used for 10 minutes at 45-50°C followed

by centrifugation clarifies the sap. Diffusion occurs at the interface and a disc of precipitation (ring) occurs where reactants have reached optimal proportions. Interaction of virus sample (antigen) and virus specific antiserum to produce a precipitate indicates a relationship between the unknown virus and the virus used in antiserum production. For most diagnostic requirements, a tube-precipitation or micro-drop test for filamentous viruses is quite satisfactory.

- (c) Microprecipitin test :** Antibody (Ab) is diluted with physiological saline (0.85% NaCl) in water or buffer and so also the antigen (Ag) with appropriate buffer in serial dilution of 1:10, 1:20, 1:40, 1:80 and 1:160. A drop of each Ag dilution is placed per well of microtitre (haemagglutination) plastic tray of each column. Then a drop of each Ab dilution is added per well of each row. The two drops are then stirred together in each well. Drops in the trays are usually covered with liquid paraffin to prevent evaporation.

Cloudy precipitate forms in drops with proper dilutions of homologous Ag and Ab within hours which can be revealed by observing the drops under the microscope indicating a relationship between the unknown virus (antigen) and the virus used in antiserum production.

B. Precipitin tests in agar (Gel diffusion test)

(a) Ouchterlony double diffusion test : In the double diffusion test, high grade agar (0.9% agarose or ion agar No.2) is prepared in buffer (pH 7.6) containing 0.85% NaCl in a preservative such as 0.02% sodium azide. The agar is poured into plastic Petri dishes to a depth of 2-3 mm or on to glass slides to a depth of 1 mm. Wells are cut in the set agar, either with a special cutter or with a cork borer. Virus samples are placed in the center well and antiserum in the outer wells and vice-versa. The plates are incubated in a moist atmosphere at constant temperature. Viruses diffuse in a ring from the centre well and the antiserum from the outer wells. If the virus has antigenic sites in common with the antiserum, then a precipitate (band or spur) will form in the agar and this becomes visible between the two wells. Serologically identical or very closely related viruses shows bands whereas distantly related viruses form spurs (a short projecting part of any kinds). Unrelated viruses / healthy sap show no precipitation.

Various patterns of precipitation lines can be obtained depending on the relationship between the antigen and antibody. Because elongated viruses diffuse rather poorly through agar, the techniques may be modified either to use lower concentration of agar (0.5%) or the virus must be disrupted to produce small more diffusible fragments, or otherwise tube (ring interface) precipitation is best to be used for elongated viruses. However these immune-diffusion tests in agar gel are entirely satisfactory for isometric viruses.

(b) Radial diffusion test : In this method antiserum or antigen can be incorporated into agar. Wells are cut in the agar to contain the antigen; diffusion occurs and where the concentration of the reagent reaches the right proportions, a precipitate is formed. To test *Barley stripe mosaic virus* (BSMV), antiserum to BSMV (a rod-shaped virus) is incorporated into agar, together with a virus dissociating agent, in 90 mm Petri dishes. Small samples of cereal leaves under test are inserted into the set agar about 1 mm apart in rows 2 mm apart; in this way 400 samples can be tested. The plates are incubated for 24-36 hours and then viewed with a

binocular microscope for the presence of a white precipitate. This test can be used to assess cereal grains for virus; it can detect 1µg of BSMV per ml, approximately a 10-fold greater sensitivity than double-diffusion tests.

C. Chloroplast agglutination test :

This test can be performed on crude sap extracts of suspected virus-infected plants. Plant material is ground to produce a sap suspension of chloroplasts and cell debris. Antiserum to the suspected virus is added and the mixture left for a few minutes on the slide. If virus is present, cloudy chloroplast agglutination is visible in the liquid where the antibody and the matching virus meet and adhere together.

D. Latex agglutination test

Antiserum (Ab) is adsorbed on to polystyrene latex beads and then mixed with crude sap extracts of suspected virus-infected plants (antigen) in a round bottom micro titer plates and left for a few minutes. Flocculation of the latex indicates positive reaction of antibody-antigen combination. This allows the detection of 100-1000 fold smaller quantities of virus present in the crude extract within 5-10 minutes. The technique is more sensitive than other precipitation tests.

When *antigen concentration is low* precipitation does not become visible and in such cases *indirect tests are performed* to find out the combination between antigen and antibody.

Figures:

Fig.1- Tube precipitin test

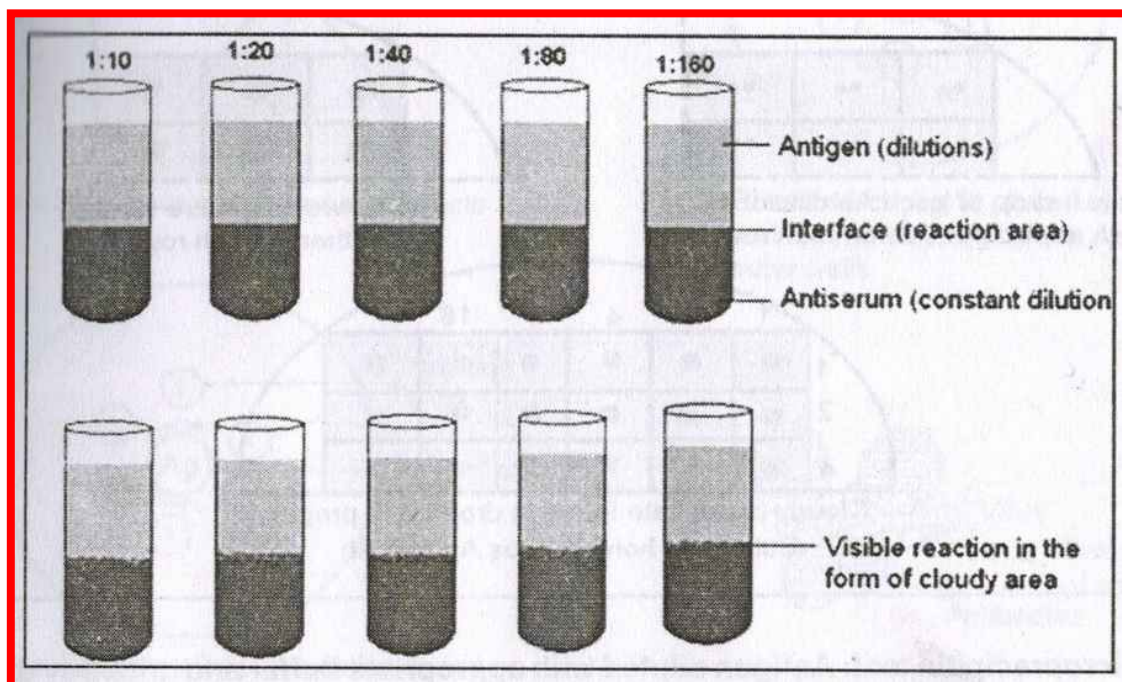


Fig2: Microprecipitin test

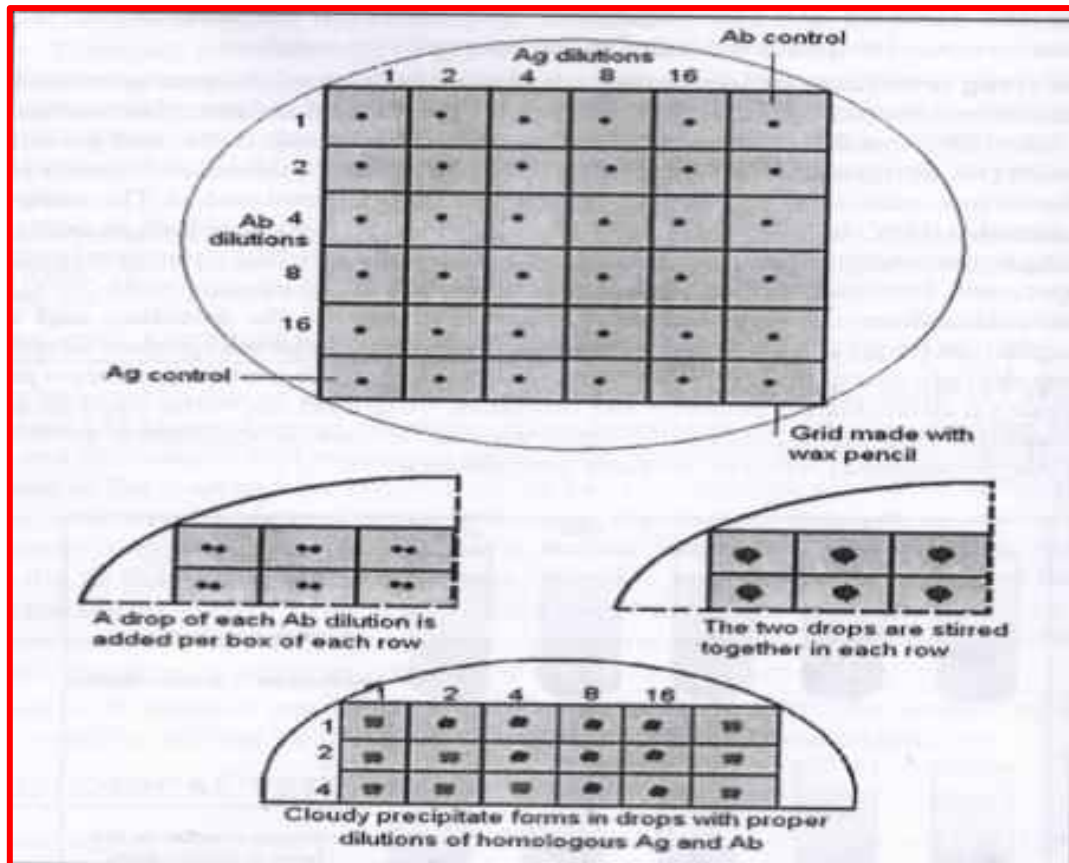


Fig2: Ochterlony double diffusion test

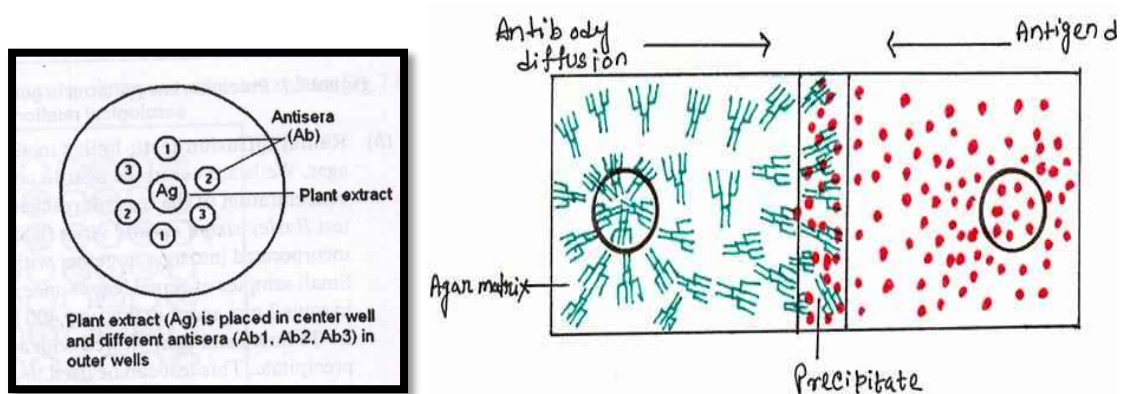


Fig.4 Radial diffusion test

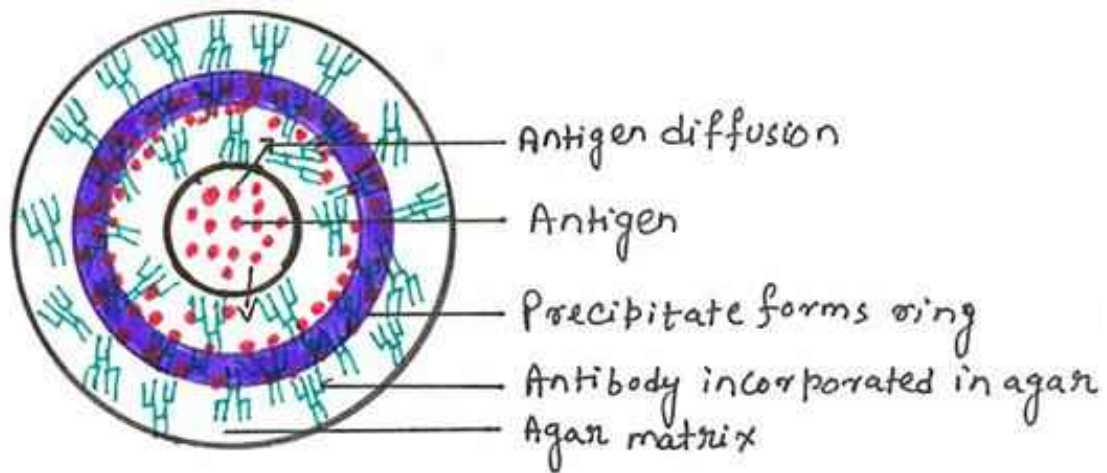


Fig 5. Latex agglutination test

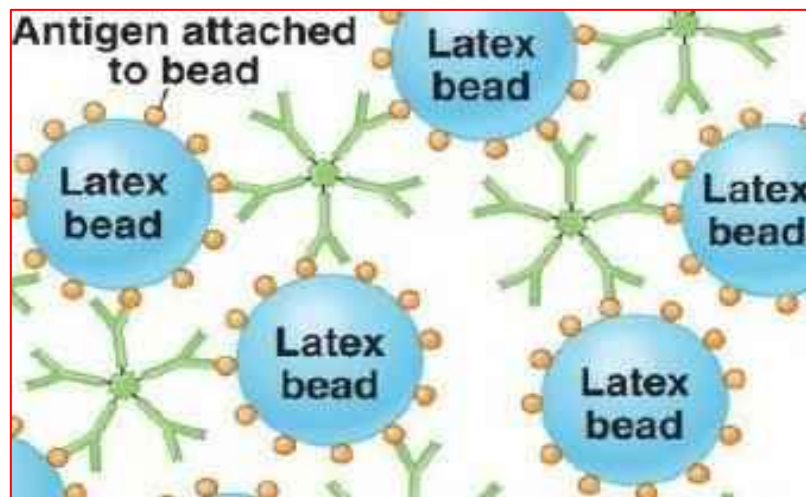
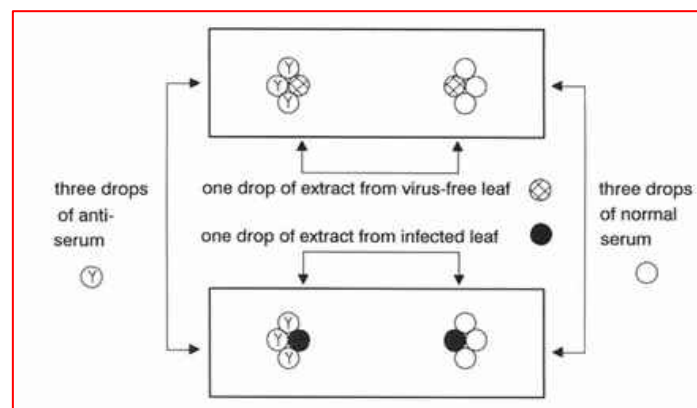


Fig 6. Chloroplast agglutination test :



3. Enzyme-linked immunosorbent assay (ELISA) :

Precipitin and agglutination tests require that enough antibody and virus be present to make a visible agglutination or precipitation i.e. these tests are less sensitive. But in the ELISA test, the substrate begins to turn colour due to the presence of the enzyme, and even very low virus titers are detectable. Virus concentration also can be quantified by measuring the colour intensity. ELISA allows rapid testing of numerous plants using only small sap samples once antibodies are available.

While there are many variations of ELISA, the two types most commonly used are the double antibody sandwich assay (DAS/Direct-ELISA) and indirect ELISA (I-ELISA).

A. DAS-ELISA

The DAS-ELISA test is performed as follows :

1. Antiserum against a particular virus of interest is treated with ammonium sulphate to precipitate the g—globulin antibodies.
2. Precipitated g—globulin antibodies are subject to dialysis and are suspended in buffer.
3. A portion of antibody suspended in buffer is then conjugated to an enzyme such as alkaline phosphatase ((β -galactosidase).
4. The test is then performed in 96-well flat-bottom polystyrene microtiter plate.
5. Antibodies un-conjugated ones specific for a particular virus, are first added to wells in the plate to coat the well surfaces.
6. After incubation unabsorbed antibodies are washed from the wells and sap samples containing the virus (antigens) are added, one sample per well.
7. If the virus is present in the sap, it will react with the adsorbed antibodies and thus be bound to the plate. The excess antigen is washed from the wells.
8. Next after washing, each well is treated with a solution containing antibody-enzyme conjugate. Wherever the virus is bound in a well, the enzyme will also be bound by means of a second serological reaction. It means if the expected virus is present; the second antibody also fits onto the virus, completing the "sandwich" and remains in the well, while excess material is washed away. Non-specific binding of enzyme is prevented by special buffer components.
9. The wells are washed free of unbound conjugate
10. Finally the wells are filled with p-nitro-phenyl phosphate substrate upon which the enzyme in the conjugate can work. The enzyme activity is measured after incubation on the basis that the amount of hydrolysis is directly proportional to the amount of virus present. In plant virus research, alkaline phosphatase is generally used as the enzyme of choice because of its colourless substrate p-nitrophenyl phosphate.
11. Conversion of the substrate to a yellow coloured product due to the degradation of enzyme substrate and production of p-nitro-phenol indicates a positive result. The end product of the reaction, p-nitrophenyl, is yellow in colour and so can be measured quantitatively. The colour not only shows that antigen (virus) is present in the plant extract but also gives a quantitative measure of virus in each sample, since the amount of colour produced depends on the amount of enzyme in each well and this in turn depends upon the

degree of antigen-antibody binding. The change in the colour of the end product are recorded by scanning the optical density of each well of microtiter plate with a multi-scanner spectrophotometer called the ELISA plate reader

During the test, it is important that negative and positive control samples must be incorporated into each test to enable distinction between healthy and diseased plants and to serve as a control for reagent and test conditions. Positive controls may consist of fresh sap from plants infected with the virus (es) in question, and maintained in the glasshouse for that purpose.

B. Indirect ELISA (I-ELISA)

In I-ELISA, antibody is bound to a solid support, as in DAS-ELISA, but binding of the antigen is recognized by an antibody raised in a second animal. (The second antibody is not conjugated with an enzyme but it is in turn recognized by a further antibody which is conjugated to an appropriate enzyme)

In this method, microtitre wells are first treated with antigen and then with antiserum (antibody) produced in rabbits. The enzyme, however, is conjugated to globulins that recognize rabbit serum. This anti-rabbit globulin with its attached enzyme is added to the microtitre wells and then the substrate added. This method lowers the specificity of the test because there is likely to be more complete binding of the un-conjugated antibody and for this reason it may be less strain specific than DAS-ELISA. However, it allows the detection of wider range of serologically related viruses. This test is not suitable for crude extract since substances in plant sap inhibit adsorption of the virus to the microtitre plates.

C. Competitive ELISA

In competitive ELISA the steps are somewhat different than the other two ELISA methods. In this method free antigen and antibody (unlabeled) are first incubated to form antigen-antibody complex and then these bound antigen/antibody complexes are added to an antigen-coated well. The plate is washed with buffer to remove unbound antibody (The more antigen in the sample, the less antibody will be able to bind to the antigen in the well, hence *competition*) before adding enzyme-linked secondary antibody, specific to primary antibody. Finally substrate is added. The antigen concentration can then be determined by the signal strength elicited by the enzyme-substrate reaction. In this assay, the enzyme-linked secondary antibody competes with the sample antigen which is associated with the primary antibody, hence known as competitive ELISA. In competitive ELISA crude or impure samples can be used and still selectively binds any antigen that may be present.

D. Blot ELISA

i.Dot blots Immunobinding Assay (DIBA) : A drop of a test sample (crude sap) is added on the nitrocellulose membranes containing specific monoclonal antibodies

ii.Dip sticks ELISA : Nitrocellulose dip stick surface coated with a specific monoclonal antibodies are dipped into the test sample (crude sap). The antigens are detected directly by enzyme labeled virus specific antibodies (**Direct ELISA**)

iii.Tissue blot immunoassay : Infected freshly cut plant tissue is directly electro-blotted on to the nitrocellulose membrane (transfer of viral antigen to membrane.) The

antigens are reacted first with primary antibodies specific to the virus then detected with enzyme labeled secondary antibodies that react with primary antibodies (**Indirect ELISA**)

The presence of antigen (virus) is detected following incubation with the antibody enzyme conjugate, washing with buffer and finally by addition of the enzyme substrate. For the final colour development a substrate is added. The enzyme linked to the IgG converts to an insoluble coloured material. The intensity of the coloured spot can be assessed by eye or by using a reflectance densitometer.

Blot ELISA is helpful in detecting a large no. of samples in the field as nitrocellulose membrane is easy to transport. Membrane blots after being prepared can be stored for a period of up to 4 weeks in a dust free environment and sent to another laboratory for detection.

4. Molecular diagnosis

Nucleic Acid Based Detection Techniques

1. Polymerase Chain Reaction (PCR)
2. Southern Blot
3. Northern Blot
4. Microarray Technology

1. Polymerase Chain Reaction (PCR)

- **Polymerase Chain Reaction** is an *in vitro* technique for the amplification of a **specific sequence of DNA which** is used for further testing.
- PCR was discovered by Kary Mullis in 1985 for which he was awarded the Nobel Prize in 1993.
- PCR targets and amplifies a specific region of a DNA strand.
- It is an *in vitro* technique to generate large quantities of a specified DNA.

PCR consists of 3 steps-

- **Denaturation** at high temperature (94⁰ C) to separate the 2 complementary strands of dsDNA.
- **Annealing** of two oligonucleotide primers to their complementary sequences in the opposite strands of the target DNA
- **Extension** of each primer through the target region (usually at 72 °C) using a thermostable DNA polymerase (e.g., *Taq* polymerase).

Temperature Protocol

- Initial Melt: 94°C for 2 minutes
- Melt: 94°C for 30 seconds
- Annealing: 55°C for 30 seconds
- Extension : 72°C for 1 minute
- Final Extension: 72°C for 6 minutes
- Hold: 4°C

Types and Modifications of PCR

- Reverse Transcriptase PCR
- Real Time PCR
- Allele Specific PCR
- Multiplex PCR
- Nested PCR

2. Southern blotting

Technique that is widely used to detect fragments of DNA.

1. Electrophoresis.
2. Denatured with alkali.
3. DNA fragments are blotted onto a strip of nitrocellulose or a nylon membrane.
4. Identification by hybridization with a labeled, complementary nucleic acid probe.

3. Northern blotting

Technique for detection of specific RNA.

1. RNA sample are more susceptible to degradation than DNA.
2. We separate RNA molecules by size using agarose gel electrophoresis
3. RNA is blotted on to a nylon positively
4. The membrane is placed in a hybridization buffer with a charged membrane
5. Labeled probe is detected by autoradiography.
6. The gene expressed will show a signal band upon detection.

4. MICROARRAY TECHNOLOGY

- Microarrays are generally composed of thousands of specific probes spotted onto a solid surface (usually nylon or glass).
- Each probe is complementary to a specific DNA sequence and hybridisation with the labelled complementary sequence provides a signal that can be detected and analysed.

PREPARATION OF PROBE:

- PCR fragments arrayed on nylon membranes, hybridised against cDNA samples radioactively labelled, called macroarrays
 - PCR products spotted onto glass slides and DNA labelled with fluorescent dyes
 - Oligonucleotides of different length (from 18 to 70 bp) arrayed and hybridised with the same type of labelled DNA material.
-

Lec.7. Avoidance and exclusion- plant quarantine – domestic, international and embargo- phytosanitary certificate- quarantine in India. Post entry quarantine in India. Exotic diseases introduced in India.

Avoidance

- It involves tactics that prevent contact between the host and the pathogen.
- The selection of geographic area, selection of a proper field, planting time and disease escaping varieties play an important role in avoiding the disease.

For several plant diseases, control depends on attempts to evade pathogens.

- Bean anthracnose, caused by the fungus *Colletotrichum lindemuthianum*, and the bacterial blight of bean caused by bacteria *Xanthomonas phaseoli* and *Pseudomonas phaseolicola* are transmitted through the seed. Therefore, they can be successfully controlled by using disease free seed and seed treatments.
- In many cases, the susceptible crop is planted at a great enough distance from field containing infected plants so that the pathogen would not infect the crop.
- Crop isolation is practiced mostly with perennial plants, such as peach orchards isolated from choke cherry shrubs or trees infected with X disease phytoplasma.
- Various activities which evade the pathogens include:
 - i) Using vigorous seed
 - ii) Selecting proper dates and proper sites
 - iii) Maintaining proper distances between fields and between rows and plants
 - iv) Planting windbreaks or trap crops
 - v) Planting in well drained soil
 - vi) Using proper insect and weed control

Such practices increase the chances that the host will remain free of pathogen or at least that it will go through its most susceptible stage before the pathogen reaches the host.

Use of Pathogen Free Seed and Propagative Material

- Seed may carry internally one or a few fungi such as those causing anthracnose and smuts, certain bacteria causing bacterial wilts, spots and blights and certain viruses (*Tobacco ring spot virus in soybean*, *Bean common mosaic virus*, *Lettuce mosaic virus*, *Barley stripe mosaic virus*, *Squash mosaic virus* and *Prunus necrotic ring virus*). Such diseases can be controlled effectively by producing and using disease free seed.
- True seed, however, is invaded by relatively few pathogens, although several may contaminate its surface.

- All types of pathogen can be carried in or on propagating material.
- When a pathogen is excluded from the propagating material of the host, it is often possible to grow the host free of that pathogen for the rest of its life, e.g., woody plants, generally affected by non-vectored viruses.

Production of pathogen free vegetative propagating material

- Vegetative propagating material free of pathogens that are distributed systemically throughout the plant is obtained from mother plant that had been tested and shown to be free of particular pathogen or pathogens.
- To ensure continuous production of pathogen free buds, grafts, cuttings, rootstocks and runners of trees, vines, and other perennials; the mother plant is indexed for the particular pathogen at regular intervals.
- For certain crops, such as potato, complex certification programmes have been evolved to produce pathogen free seed potatoes.
- For the seed to be certified the plants must show disease level no higher than those allowed by particular state.
- Sometimes it is impossible to find even a single plant of variety that is free of particular pathogen, especially of viruses. In that case, one or few healthy plants are initially obtained by meristematic tissue culture which most viruses do not invade.

Practices for the Exclusion of Pathogens from Plant Surfaces by Epidermal Coatings

- The plants are sprayed with compounds that form a continuous film or membrane on the plant surface for controlling diseases of aboveground parts of plant and inhibit contact of pathogen with the host and penetration of host.
- Water emulsion of dodecyl alcohol forms a high quality of lipid membrane. The membrane allows diffusion of oxygen and carbon dioxide but not of water. The membrane is not easily washed by rain and remains intact for about 15 days.

Kaolin based films have also proved effective in protecting apple shoot from becoming infected by the bacterial disease fire blight, and apple fruit from powdery mildew. It also protects grapevine from Pierce disease caused by *Xylella fastidiosa* by interfering with its transmission by the vector.

Exclusion

- It means preventing the entrance and establishment of pathogens in uninfected crops in a particular area.
- It can be achieved using certified seed or plants, sorting bulbs before planting, discarding any that are doubtful, possibly treating seeds, tubers or corms before they are planted and most importantly refusing obviously diseased specimens from dealers.

- In order to prevent the import and spread of plant pathogens into the country or individual states, certain federal and state laws regulate the conditions under which certain crops may be grown and distributed between states and countries.
- Such regulatory control is applied by means of quarantine, inspection of plants in the field or warehouse and occasionally by voluntary or compulsory eradication of certain host plants.
- Plant quarantines are carried out by experienced inspectors, stationed in all points of entry into the country, to stop persons or produce likely to introduce new pathogens.
- Similar quarantine regulations govern the interstate and even intrastate sale of nursery stock, tubers, bulbs, seeds and other propagative organs, especially of certain crops such as potatoes and fruit trees.
- For example, the outbreak of citrus canker in USA in 1910 through planting material imported from Southeast Asian countries.
- Due to heavy destruction, strict quarantine was imposed against entry of citrus planting material.
- However in 1981, 1984 and 1991, fresh outbreaks were reported due to illegal importation of citrus planting material. In India, interstate quarantine is in place for the movement of potato from Darjeeling area of West Bengal to prevent the spread of potato wart which is restricted to that area only.

Plant Quarantine

The term '**Quarantine**' means simply forty i.e., 40 days period. This was more commonly referred to the period of detention for ships arriving from countries subject to epidemic diseases such as the Bubonic plague, cholera and yellow fever. The crew and the passengers used to be compelled to remain isolated on board for sufficient period to permit the diseases to develop and be detected. The purpose of the health authorities was to establish adequate detention period. Later on, the term 'Quarantine' came to be only used for the detention and the practices connected with it. The term got associated from the human disease field to the animal disease field and later on adopted to cover protective methods for the exclusion of pests and diseases of agricultural and horticultural crops.

In strict sense '**Plant Quarantine**' refers to the holding of plants in isolation until they are believed to be healthy. Now, broader meaning of the plant quarantine covers all aspects of the regulation of the movement of living plants, living plant parts/plant products between politically defined territories or ecologically distinct parts of them. Intermediate quarantine and post entry quarantine are used respectively to denote the detention of plants in isolation for inspection during or after arrival at their final destination.

Importance

The entry of a single exotic insect or disease and its establishment in the new environment continues to cause great, national loss (table) till such time it is brought under effective control. In certain cases a country has to spend a few million rupees before success in controlling the introduced insect pest or disease is achieved.

History

The first plant quarantine law was promulgated in Rollen, France in 1860 to suppress and prevent the spread of common barberry, the alternate host for wheat stem rust. Among other countries, the first few to establish plant quarantine services were Germany, France, Australia and the U.S.A. In India, legislative measures against crop pests and diseases was initiated under the Destructive Insects and pests Act of 1914 (DIP act) and it was passed by Governor General of India on 3rd February, 1914. Under this Act, rules governing the import and movement of plants and plant materials, insects and fungi are framed. The Act provides

1. It authorizes the Central Government to prohibit or regulate the import into India or any part there of any specific place therein, of any article of class of articles.
2. It authorizes the officers of the Customs at every port to operate, as if the rules under the D.I.P. Act is made under the Sea Customs Act.
3. It authorizes the Central Government to prohibit or regulate the export from a State of the transport from one State to another State in India of any plants and plant materials, diseases or insects likely to cause or infestation. It also authorizes the control of transport and carriage and gives power to prescribe the nature of documents to accompany such plants and plant materials and articles.
4. It authorizes the State Governments to make rules for the detention, inspection, disinfection or destruction of any insect or class of insects or of any article or class of articles, in respect of which the Central Government have issued notifications. It also authorizes the State Governments for regulating the powers and duties of the officers whom it may appoint on this behalf.
5. It provides penalty for persons who knowingly contravene the rules and regulations issued under the Act.
6. It also protects the persons from any suit or prosecution or other legal proceedings for anything done in good faith or intended to be done under the Act.

Consequent to Bengal famine 1943, a Central Plant Protection organization was established in 1946 under the then Ministry of Food and Agriculture.

Often a new pest, disease or weed has accidentally entered a country where it did not exist before and has multiplied, spread and caused enormous damage to the crops of that country. For instance powdery mildew of grapevine (*Plasmopara viticola*), introduced into France from America, was responsible for the destruction of the vine industry of that country until hybridization with resistant American stock offered a solution. The blight disease of chestnut

(*Endothia parasitica*) which was introduced into U.S.A. from Asia in 1904, completely wiped out chestnut trees.

Coffee rust (*Hemileia vastarix*) which came into India in 1879 from Sri Lanka is now widespread in all coffee growing areas. Fire blight (*Erwinia amylovora*) of pear and other pomes which was introduced from England in 1940 is well established in Uttar Pradesh. Late blight (*Phytophthora infestans*) of potato introduced into India in 1889 from Europe is now present in many parts of the country. Flag smut (*Urocystis tritici*) of wheat introduced from Australia is now well spread in Madhya Pradesh, Punjab, Rajasthan and Uttar Pradesh. Rubber powdery mildew (*Oidium heavea*), which was introduced from Malaysia in 1938, is also causing great concern in Kerala. Black rot of crucifers (*Xanthomonas campestris* pv. *campestris*) believed to have been introduced to India with seeds imported from Holland, and other European countries after World War II, prevailed for some years on the hills and then spread to the plains and became established in Indian seed stocks, especially in West Bengal.

Among the more important plant disease introductions, mention may be made of bunchy top virus of banana introduced from Sri Lanka in 1940 which has since spread widely in Kerala, Orissa, West Bengal and Assam. The wart disease (*Synchytrium endobioticum*) of potato was first noticed in Darjeeling district of West Bengal having been introduced with seed potatoes from Holland. By 1962, the disease spread over nearly 1000 ha and has recently been reported from Nepal also. The mosaic disease of banana is another introduced disease which is only confined to Gujarat and Maharashtra states. Recently the apple scab (*Venturia inaequalis*) which was only confined to small area in Jammu and Kashmir has now appeared in severe form in many locations in Himachal Pradesh, and is posing a problem to apple industry.

The establishment of a plant quarantine regulation should rest on the following fundamental pre-requisites.

- i. the pest/disease under consideration must be one that will offer actual or expected threats to substantial interests (Agricultural and / or commercial)
- ii. the quarantine regulation or degree must represent a measure for which no substitute action involving less interference with normal activities is available.

Diseases believed to have been introduced into India from foreign countries

Disease	Host	Date of first record	Introduction from
1. Leaf rust (<i>Hemileia vastarix</i>)	Coffee	1879	<i>Sri Lanka</i>
2. Late blight (<i>Phytophthora</i>	Potato, Tomato	1883	Europe

<i>infestans</i>)			
3. Rust (<i>Puccinia carthami</i>)	Chrysanthemum	1904	Japan or Europe
4. Flag smut (<i>Urocystis tritici</i>)	Wheat	1906	Australia
5. Downy mildew (<i>Plasmopara viticola</i>)	Grapevine	1910	Europe
6. Downy mildew (<i>Pseudoperonospora cubensis</i>)	Cucurbits	1910	Sri Lanka
7. Downy mildew (<i>Sclerospora philippinensis</i>)	Maize	1912	Java
8. Foot rot (<i>Fusarium moniliforme</i> var. <i>majus</i>)	Rice	1930	South East Asia
9. Leaf spot (<i>Phyllachora sorghi</i>)	Sorghum	1934	South Africa
10. Powdery mildew (<i>Oidium heveae</i>)	Rubber	1938	Malaya
11. Black shank (<i>Phytophthora parasitica</i> var. <i>nicotianae</i>)	Tobacco	1938	Dutch East Indies
12. Fire blight (<i>Erwinia amylovora</i>)	Pear and other pomes	1940	England
13. crown-gall and hairy root (<i>Agrobacterium tumefaciens</i> A. <i>rhizogenes</i>)	Apple, Pear	1940	England
14. Bunchy top	Banana	1940	Sri Lanka
15. Canker (<i>Sphaeropsis malorum</i>)	Apple	1943	Australia
16. Wart (<i>Synchytrium endobioticum</i>)	Potato	1953	Netherlands

Despite every precaution of inspection, certification and treatment, it is not always possible to guarantee that a consignment is completely free from pathogens. In doubtful cases it is advisable to subject plants to a period of growth in isolation under strict supervision in the importing country (post-entry quarantine). The plants are grown at a quarantine station. When direct importation of plants to a country's own quarantine station is considered very dangerous, quarantine during transit from the country of origin (intermediate quarantine) may be required. The requirements of an intermediate station are similar to those for a post-entry station. Intermediate quarantine inspection must always be followed by post-entry quarantine after arrival of the consignment at its final destination. During post-entry or intermediate quarantine plants must be kept under close supervision, so that any pest or disease which appears may be immediately detected and grown under optimum conditions, so that symptoms are not marked by physiological disturbances.

Classification of quarantine:

Quarantine can be classified as Exclusive and Regulatory

Exclusive:

Prohibit importation of plant materials into a specified area, this is resorted when no satisfactory way is found to permit the entry of plant materials.

Regulatory:

The import permitted if a satisfactory "Phytosanitary Certificate" accompanies the material.

Quarantine can also be classified as

- 1) Domestic
- 2) International
- 3) Embargos

1) Domestic:

It restricts the spread of a disease within a country, from one place to another within the state or between states.

Quarantine in Tamil Nadu:

The Madras Agriculture Pests and Diseases Act 1919 amended in 1955 provides control of sandal spike, cardamom mosaic, tobacco broomrape, wheat stem rust and tea blister blight.

Quarantine in India:

Under the DIP Act, the Directorate of Plant Protection, Quarantine and Storage has the responsibility to take the necessary steps and regulate the inter-state movement of plants and plant material in order to prevent the further spread of destructive insects and diseases that have already entered the country. The sole object of enforcing domestic quarantine is to prevent the

spread of these diseases from infected to non-infected areas. Currently, domestic plant quarantine exists in four diseases, wart (*Synchytrium endobioticum*) of potato from 1959, bunchy top (virus) of banana from 1959, mosaic (virus) of banana from 1961 and apple scab (*Venturia inaequalis*) from 1979.

Most of the states in India have plant quarantine laws to avoid entry of plant pests and diseases.

1. **Bunchy top of banana:** The export and the transport from the States of Assam, Kerala, Orissa, West Bengal, Tamil Nadu to any other State of Banana plant or any other plant of the genus *Musa*, including sucker, stem, leaf, flower, and any other part thereof which may be used for propagation, or the materials of banana plant or any other plant of the genus *Musa*, which are used for packing and wrapping, excluding the banana fruit is prohibited.

2. **Banana mosaic :** The export and transport from the States of Maharashtra and Gujarat of any plant of Banana or any other plant of genus *Musa* including the sucker, stem, flower and any other part thereof, but excluding leaf and fruit thereof is prohibited; vide Government of India notification No.F. 6-10-PPS dated the 11th April, 1961.

3. **Potato wart:** The export to potato tubers from the State of West Bengal to any other State or territory of India is prohibited.

4. **Apple scab:** The Directorate of Horticulture, Himachal Pradesh worked out a detailed scheme for the eradication of scab, and also issued a notification No.NIC.20/76 dated 28 December 1978, prohibiting the export of planting material of apple outside the State.

2) **International (foreign):**

The Foreign quarantine deals with the movement of plants/plant materials the countries in order to ward off the risk of the infiltration of exotic pests and diseases into the countries. This order prescribes regulations for the import of plant/plant materials.

3. **Embargo:**

Total prohibition on import/export of transport of plant material in a country.

Post –entry quarantine (PEQ)

Plants and seeds, which require post-entry quarantine shall be grown in post-entry Quarantine facilities, approved and certified by the Designated Inspection Authority, to confirm to the conditions laid down by the Plant Protection Adviser. The period for which, and the conditions under which, the plants and seeds shall be grown in such facilities shall be specified by the plant protection adviser.

National Bureau of plant Genetic Resources (NBPGR) is the nodal agency for the exchange of germplasm material of all agri-horticultural and silvi-agriculture crops for research purposes by the institutions under the ICAR system and private organization. NBPGR has been authorized to all import of germplasm materials by authorized ICAR institutes and private agencies engaged in research work in accordance with the provisions of PFS order, 1989.

Plant Quarantine Methods

- 1) prohibition
- 2) interception
- 3) elimination

Prohibition:

It takes the form of embargoes characterized as quarantines. Embargoes are absolute prohibitions directed at pest/pathogens of such potentially high risk that plants or other items capable of harboring those pests/pathogens are prohibited from entry without exception.

Interception:

It stops pathogens at designation points and is accomplished by inspection of subject items. If inspection detects the pathogens the item is not admitted. All countries enforcing quarantines have some manner of any inspection program. Inspections can be general or specific. During a general inspection any plant material, soil, containers, or similar items are examined for any pathogen. The inspections are intended to intercept plant disease of unknown or questionable economic importance that is not present or widely distributed within the country. A specific inspection consists of searches for economically important plant diseases that are not present or widely distributed. The technology involved in quarantine inspections can be as simple as a hand lens or as advanced as a scanning electron microscope.

Elimination:

It involves various techniques to produce or ensure diseases free material at its source, in contrast to cleaning it up at its destination. Elimination cannot be separated from prohibition and interception since the latter strategies often require elimination by pre entry or post entry treatment using chemicals or heat to eliminate pathogens. Elimination is also broader in scope than prohibition and interception.

Quarantine procedure:

- It is essential that the propagules must be clean, healthy and free from weeds and insect pests. They must not be treated by the sender with fungicides or insecticides.
- Each imported entry or sample must be accompanied by a phytosanitary certificate from the scientist/ institution sending the sample.
- If the sample is contaminated, the materials is valuable, the contaminated entries may be grown in isolation in an effort to isolate some healthy plants.
- Samples/entries that are heavily contaminated are destroyed by the bureau.

Restriction on export and introduction of plant materials:

Introduction of certain plant species in India is restricted. Cotton, berseem, linseed and sugarcane seeds cannot be obtained through letter post.

Introduction of following is prohibited: cocoa from Africa, Ceylon and West Indies, sugarcane from Australia, Fiji, New Guinea, Philippines and West Indies, rubber from America and West Indies and sunflower from Argentina and Peru. Entry of unginned cotton and of Mexican

jumping beans is strictly prohibited. India restricts, more often prohibits, the export of propagules of tea, jute and black pepper.

International plant protection convention

The first effort towards international agreement on Plant Protection was made in 1914 under the auspices of the International Institute of Agriculture in Rome. This was followed by an International Convention of Plant Protection by over 50 member countries of the Institute in 1919 and certain Agreements regarding the issue and acceptance of phytosanitary certificates were finalized. The project received a set back due to Second World War and was later on revived by the FAO.

In post-war period International action in Plant Protection and particularly in plant quarantine was encouraged by FAO with the establishment in 1951 of the International Plant Protection Convention. This agreement was constituted with the purpose of securing common and effective action to prevent the introduction and spread of pests and diseases of plants and plant products as to encourage Governments to take all steps necessary to implement its prevention (Ling, 1953).

The following regional Plant Protection Organizations are now in operation.

1. The European and Mediterranean Plant Protection Organization (EPPO)
2. The Inter-African Phytosanitary Council (IAPSC)
3. Organismo Internacional Regional de Sanidad Agropecuario (OIRSA)
4. The Plant Protection Committee for, the South East Asia and Pacific region.
5. Comit'e Interamericano de Protection Agricola. (CIPA)
6. The Caribbean Plant Protection Commission (CPPC)
7. The North American Plant Protection Organization (NAPPO).

Under article 3 of that International Plant Protection Convention, the Plant Protection Agreement for South East Asia and Pacific Region was sponsored by F.A.O in 1956, and India became in party to this Agreement in the same year the along with Australia, Sri Lanka, the U.K., Laos, Netherlands, Indonesia, Portugal and Vietnam. Our Government agreed to adopt legislative measures specified in the Convention for the purpose of securing common and effective action to prevent the introduction and spread of pests and diseases of plants and plant products and to promote measures for their control and also agreed to assume all responsibilities for the fulfillment within its territories of all requirements under the Convention. It was agreed that the Government shall make provision for:

Plant quarantine organizations in India

The first recorded plant quarantine measure in India dates back to 1906 when, perceiving the danger of introducing the Mexican boll weevil, the Government of India directed that all cotton imported from the New World should only be admitted to India after fumigation with carbon disulphide at the port of entry.

In India two categories of regulatory measures are in operation for controlling pests, diseases and weeds. In the first category regulatory measures are aimed to prevent the

introduction of exotic pests and diseases into the country or their spread from one State or Union Territory to another (Plant Quarantine). The second pertains to suppression or prevention of spread of pests and diseases in localized areas within a State or Union Territory. The former derives its authority from the **Destructive Insects and Pests (DIP) Act 1914** of the Central Government and the latter from Agricultural Pests and Diseases Acts of the various States.

The legislative measures against crop pests and diseases were initiated under the DIP Act of 1914 which was passed by the then Governor General of India in Council on 3 February 1914. Prior to the establishment of the Directorate of Plant Protection, Quarantine and Storage in 1946, under the Ministry of Food and Agriculture, the various rules and regulations of the DIP Act were enforced by the customs department. The quarantine regulations are operative through The Destructive Insects and Pests Act, 1914 (which has been revised 8 times from 1930 to 1956 and amended in 1967 and 1992).

The provisions of the DIP Act are

1. It authorizes the Central Government to prohibit or regulate the import into India or any part thereof or any specific place therein of any article or class of articles.
2. It authorizes the officers of the Customs at every port to operate, as if the rules under DIP Act are made under the Sea Customs Act.
3. It authorizes the Central Government to prohibit or regulate the export from a State or the transport from one State to another State in India of any plants and plant material, diseases or insects, likely to cause infection or infestation. It also authorizes the control of transport and carriage and gives power to prescribe the nature of documents to accompany such plants and plant materials and articles.
4. It authorizes the State Governments to make rules for the detention, inspection, disinfection or destruction of any insect or class of insects or any article or class of articles, in respect of which the Central Government has issued notification. It also authorizes the State Governments for regulating the powers and duties of the officers whom it may appoint on its behalf.
5. It provides penalty for persons who knowingly contravene the rules and regulations issued under the Act.
6. It also protects the personnel from any suit or prosecution or other legal proceedings for anything done in good faith as intended to be done under this Act.

The quarantine regulations are operative through “The Destructive Insects and Pests Act, 1914 (which has been revised and time from 1930 to 1956 and amended in 1967 and 1992).

The Act also empowers the State Governments to frame suitable rules and issue notifications for inter-state movement of plant and plant material. Those rules are known as plant quarantine rules.

Under the Act, Central Government frames rules prescribing the seaports, airports and land frontiers through which plants and specified plant material can enter India, and the manner

in which these can be imported. The DIP Act operates under the National Sea Customs Act and the points of entry are located within the jurisdiction of State on the advice of Central Government, the State frames rules for detention, inspection, disinfection and destruction (as against entry) of material, if required, and delegates powers in this regard to concerned authorities with the enforcement of rules.

The plant quarantine service is centrally organized and administered through the Directorate of Plant Protection, Quarantine and Storage established under the Ministry of Agriculture (Department of Agriculture and Co-operation) which is headed by the Plant Protection Adviser to the Government of India and having its headquarters at N.H. IV, Faridabad, Haryana State.

Import regulations

When plants are imported the following principles should be followed. Some plant pathogens and pests are generally distributed in most parts of the world but others are more or less restricted in their occurrence. In some cases this limitation is due to such factors as unsuitable environmental conditions or lack of the required host plant, but in many other cases the absence of a pathogen. Most countries are aware of the desirability of delaying for as long as possible the arrival of exotic pathogens and take action to prevent their spread by introducing legislation and setting up organizations to prevent their entry. Plant quarantine legislation varies from country to country but in most cases it restricts or prohibits the importation of the pests or pathogens themselves, plants on which they might be living, soil which might be infested, foodstuffs which might carry them, and packing materials, particularly those of plant origin. Good legislation is as brief and clear as possible, at the same time being easy to interpret, gives adequate protection without interfering more than is essential with trade, and contains only restrictions which are scientifically justifiable.

When plants are imported there are certain principles which, if followed ensure that as few risks as possible are taken.

1. Import from a country where, for the crop in question, pathogens which are particularly to be guarded against are absent.
2. Import from a country with an efficient plant quarantine service, so that inspection and treatment of planting material before despatch will be thorough, thus reducing the likelihood of contaminated plants being received.
3. Obtain planting material from the safest known source within the selected country.
4. Obtain an official certificate of freedom from pests and diseases from the exporting country. Treatment of the material in the country of origin may be done; this should be noted on the certificate.
5. The smaller the amount the less the chance of its carrying infection, and inspection as well as post-entry quarantine.
6. Inspect material carefully on arrival and treat (dust, spray, fumigate, heat treat) as necessary.

7. Import the safest type of planting material, e.g. seeds are usually safer than vegetative material, unrooted cuttings than rooted. The use of axenic cultures of meristem tip tissues (micropropagation) for the international exchange of germplasm material has outstanding advantages, as such tissues can be expected to be free from latent infections by viruses, phytoplasmas etc., as well as other pathogens which are more readily detectable by visual means.
8. If other precautions are not thought to be adequate, the consignment for import should be subject to intermediate or post-entry quarantine. Such quarantine must be carried out at a properly equipped station with suitably trained staff.

Seed was not originally included in the DIP Act, but because of the changing situation and to meet the current requirements, the Government of India passed the Plants, Fruits, Seeds (Regulation of Import into India) Order 1984 which came into effect in June 1985. The conditions for the import of 17 crops are stipulated in this order. The main features of the order are:

1. Seed has been brought under the purview of the DIP Act.
2. No consignment can be imported into the country without valid import permit issued by the Plant Protection Adviser to the Government of India.
3. No consignment can be imported without an official phytosanitary certificate issued by the plant quarantine agency of the exporting country.
4. Post-entry growth of the specified crops at approved locations.

Agencies involved in plant quarantine

The authority to implement the quarantine rules and regulations framed under DIP Act rests basically with the Directorate of plant Protection, Quarantine & Storage, under the Ministry of Agriculture. This organization handles bulk import and export of seed and planting material for commercial purpose. Under this organization 9 seaports, 10 airports and 7 land frontiers are functioning. These are the recognized ports for entries for import of plant and plant material. The names and places of the ports and stations are as follows.

A. Seaports

	Place	State / Union territory
1.	Bhavnagar	Gujarat
2.	Calcutta	West Bengal
3.	Chennai	Tamil Nadu
4.	Cochin	Kerala

- | | | |
|----|---------------|----------------|
| 5. | Mumbai | Maharashtra |
| 6. | Nagapattinam | Tamil Nadu |
| 7. | Rameswaram | Tamil Nadu |
| 8. | Tuticorin | Tamil Nadu |
| 9. | Visakhapatnam | Andhra Pradesh |

B. Airports

- | | | |
|-----|-----------------|----------------|
| 1. | Amritsar | Punjab |
| 2. | Calcutta | West Bengal |
| 3. | Chennai | Tamil Nadu |
| 4. | Hyderabad | Andhra Pradesh |
| 5. | Mumbai | Maharashtra |
| 6. | New Delhi | New Delhi |
| 7. | Patna | Bihar |
| 8. | Tiruchirappalli | Tamil Nadu |
| 9. | Trivandrum | Kerala |
| 10. | Varanasi | Uttar Pradesh |

c. Land frontiers

- | | | |
|----|---------------------------|-------------|
| 1. | Amritsar Railway Station | Punjab |
| 2. | Attari Railway Station | Punjab |
| 3. | Attari-Wagah Border | Punjab |
| 4. | Bangaon Benapol Border | West Bengal |
| 5. | Gede Road Railway Station | West Bengal |
| 6. | Kalimpong | West Bengal |
| 7. | Sukhia Pokhri | West Bengal |

The Government of India has also approved three other national institutions to act as official quarantine agencies, especially for research material.

1. National Bureau of Plant Genetic Resources (NBPGR): The NBPGR in New Delhi and its regional station at Hyderabad in the agency involved in processing of germplasm, seed, plant material of agricultural, horticultural, and silvicultural crops of all the institutions of Indian Council of Agricultural Research (ICAR) functioning in the country. It is also

responsible for quarantine clearance of seed and plant material received from International Agricultural Research Centers viz., ICRISAT, ICARDA, CIMMYT, etc. ICRISAT was established in 1972 at Patancheru (near Hyderabad) to work on improvement of sorghum, pearl millet, chickpea, pigeonpea and groundnut. The quarantine clearance of all its exchanges was handled by Central Plant Protection Training Institute of Directorate of Plant Protection, Quarantine & Storage, until July 1986. This authority was later passed on to NBPGR in August 1986.

2. Forest Research Institute (FRI), Dehra Dun, for forestry plants, and
3. Botanical Survey of India (BSI) for other plants.

Quarantine inspection, treatment and certification procedures

Inspection: Inspection of plant material is an important part of plant quarantine procedure, and may be done both in the exporting country, before issue of a health certificate and after arrival to detect any pest or disease which may have become evident during transit. Publications like manuals, hand books on individual organisms of quarantine importance are prepared with illustration by each country / region to help inspectors. The following series published by Commonwealth Mycological Institute will be useful for all countries.

1. CMI descriptions of pathogenic fungi and bacteria
2. CMI/AAB descriptions of plant viruses and
3. CMI distribution maps of plant diseases.

The various steps involved in import quarantine clearance of seed and propagating plant material is outlined below

- i. Scrutinization of import application filed along with attached documents such as phytosanitary certificate (original), permit (importer's copy), shipping bill, invoice, packing list and customs bill of entry etc., to ensure the import is in order and that no prohibited plant material is imported.
- ii. Assessment of inspection fees and registration of application.
- iii. Inspection and sampling of the consignment at port warehouses or container terminal. Sampling of seed usually carried out as per the provisions of ISTA Rules and Regulations. Whereas in case of bulk import of vegetative planting material such as cuttings/saplings/budwoods/bulbs/tubers etc., at least a minimum of 0.1% of propagules are sampled variety and examined to ensure free from exotic pests or pathogens. In case of quarantine pests suspected, 100 per cent inspection is carried out for critical assessment of the risk.
- iv. Detailed laboratory testing
 - a. Visual inspection: The samples of seed/ propagating plant material is examined with the help of illuminated magnifier to record live insect infestation, contamination by soil and weed seeds, nematode galls, sclerotia, smut/bunt balls etc. Sometimes inspections are carried out under U.V. lamp to facilitate detection of specific seed-borne inspection by characteristic fluorescence.

- b. X-Ray test for detecting hidden insect infestation such as bruchids and weevils that bore into seed.
- c. Washing test to detect surface-borne oospores of downy mildew/smud spores/ bunt spores etc. and nematode cysts. Seed samples of onion, clover and lucerne are soaked for 24 to detect stem and bulb nematode and also root washings are examined for ectoparasitic nematodes.
- d. Incubation tests such as blotter test or agar plate test carried out for detecting seed-borne pathogens such as fungi. Fluorescent pseudomonas agar used for selective detection of seed-borne bacteria.
- e. Grow-out test coupled with indicator inoculation tests for detecting seed-borne viruses and bacteria.

v. Fumigation and treatment techniques: Fumigation is the versatile technique used for eliminating insect infestation. Methyl bromide is the most commonly employed for controlling insect infestation and readily adopted in quarantine programmes as the exposure time involved is short and affect all stages of insect pests and high penetrating power. Two types of fumigation viz., i. atmospheric fumigation under gas-proof sheets or chambers and ii. vacuum fumigation in vacuum chamber is widely employed. The other chemical treatments include insecticidal/fungicidal drippings or spraying or seed dressings are invariably associated with growing under post-entry quarantine conditions. The temperature treatments such as hot water treatment/ hot air treatment or vapour heat treatment are carried out to control internally borne infection/infestation and the latter particularly employed to control fruit fly infestation. Cold treatments such as refrigeration to control insect infestation in fresh fruits and vegetables. Of late, irradiation is used to control insect infestation and spoilage of food products during storage and as well as application of high intensity electronic beams through an accelerator is under experimentation.

Phytosanitary Certification

Phytosanitary or health certificate is a certificate which should accompany a plant or plant material or seed which is to be moved from one place to another place. This certificate indicates or certifies that the material under transit is free from pests or diseases.

A model phytosanitary certificate proposed at the Government consultation on the International Plant Protection convention at Rome in 1976 (Chock, 1977) and approved by F.A.O. in 1979 is given below.

MODEL PHYTOSANITARY CERTIFICATE

(to be typed or printed in block letters)

Plant Protection Organization

of _____

No. _____

To: Plant Protection Organization(s)

of _____

DESCRIPTION OF CONSIGNMENT

Name and address of exporter _____

Declared name and address of consignee _____

Number and description of packages _____

Distinguishing marks _____

Place of origin _____

Declared means of conveyance _____

Declared point of entry _____

Name of produce and quantity declared _____

Botanical name of plants _____

This is to certify that the plants or plant products described above have been inspected according to appropriate procedures and are considered to be free from quarantine pests and practically free from injurious pests; and that they are considered to conform to the current phytosanitary regulations of the importing country.

DISINFESTATION AND/OR DISINFECTION TREATMENT

Date _____

Treatment _____

Chemical (active ingredient) _____ Duration and temperature _____

Concentration _____ Additional information _____

Additional declaration:

Place of issue _____
(Stamp of Organization) Name of authorized officer _____
Date _____

(Signature)

Note: No financial liability with respect to this certificate shall attach to..... (name of plant protection organization)... or to any of its officers or representatives.

Lec 8. Eradication- Role of cultural Practices in plant Diseases management – different methods of eradication of plant diseases.

1. Eradication

Eradication is the elimination of pathogen after it has become established in the area where host is growing.

The following are the important methods followed to prevent the spread of the disease:

- i. eradication of alternate hosts,
- ii. eradication of collateral and self sown overwintering hosts
- iii. eradication of the affected plants or trees,
- iv. eradication of pathogens from infected plant parts by surgery and
- v. eradication of culled out plant materials, debris, etc., through different cultural practices

i. Eradication of alternate hosts: Removal of alternate hosts helps to prevent and check the spread of the disease caused by heteroecious rust pathogens in the primary hosts. Barberry bush is the alternate host for black stem rust pathogen *Puccinia graminis tritici* on wheat where the pathogen survives in the off-season. Barberry was eradicated in Canada, Denmark, France, Hungary, Norway and in the U.S.A. by passing stringent laws in each country. The eradication of barberry had two benefits i.e., it elimination of early spring primary inoculum and prevention of the formation of new physiologic races of the pathogens. In the U.S.A. white pine blister rust (*Cronartium ribicola*) was controlled by eradication of alternate host, *Ribes*. In Australia, Europe and the U.S.A. the apple rust (*Gymnosporangium juniperi-virginianae*) is controlled by eradication of alternate host, cedar.

ii. Eradication of collateral and self sown overwintering hosts: There are many weed hosts or wild species of cultivated plants act as collateral hosts or volunteer plants

of an economic crop which act as reservoirs of pathogens of annual crop. Reservoir hosts help the pathogen to continue the infection chain. The primary inoculum is produced on and dispersed from these hosts to the cultivated crop hosts. If these wild or uneconomic host plants of the pathogen are destroyed, the sources of primary inoculum are eliminated and chances of initiation of the disease in the crop hosts are reduced. Destruction of these hosts breaks the life cycle of the pathogen and the infection chain. Reservoir hosts or indigenous plant species which are not actually involved with the life cycle of the pathogen but provide additional sites for its persistence and multiplication. In some cases such plant species act as symptomless carriers, especially for viruses and root pathogens. Regional elimination of such hosts requires careful attention to roadside areas and other non-agricultural land also.

<i>Crop</i>	Disease	Pathogen	Collateral hosts
a. Fungi			
1. Rice	Blast	<i>Pyricularia oryzae</i>	<i>Brachiaria mutica</i> <i>Dinebra retroflexa</i> , <i>Leersia hexandra</i> , <i>Panicum repens</i> .
2. Sorghum	<i>Ergot</i>	<i>Sphacelia sorghi</i>	<i>Panicum</i> spp.
b. Bacteria			
1. Rice	Bacterial leaf blight	<i>Xanthomonas oryzae</i> pv. <i>oryzae</i>	Cyanodon dactylon, Cyperus rotundus, Leersia hexandra, <i>Leersia oryzoides</i> , <i>Panicum repens</i> , <i>Paspalum dictum</i> .
2. Apple and pear	Fire blight	<i>Erwinia amylovora</i>	<i>Hauthom</i> bushes <i>Crataegus</i> sp.
3. Cotton	Bacterial blight	<i>X.axonopodis</i> pv. <i>malvacearum</i>	<i>Eriodendron anfructuosum</i> , <i>Jatropha curcas</i> , <i>Thurbaria thespesoides</i>

c. Viruses 1. Potato 2. Bean 3. Bhendi	Rugose mosaic	Rugose mosaic virus	<i>Physalis</i> spp.
	Yellow mosaic	Bean yellow mosaic virus	<i>Sweet clover</i>
	Yellow vein mosaic	Bhendi yellow vein mosaic virus	<i>Hibiscus tetraphyllus</i>
	<i>Little leaf</i>	Phytoplasma	<i>Catharanthus roseus, Datura</i> sp.
d. Phytoplasma 1. Brinjal			

Self sown crops / volunteer plants help the pathogen to overwinter / oversummer in the absence of economic hosts. In Sudan it was enforced through legislation to pull out the cotton plants to prevent regrowth which facilitate the carry over of the cotton leaf curl virus. Wheat streak mosaic virus has been effectively controlled by eliminating the volunteer wheat plants that served as reservoirs for the virus.

iii. Eradication of affected plants or trees: In some threatening plant diseases, it is essential to eradicate the host and the pathogen from an area. Citrus canker (*Xanthomonas axonopodis* pv. *citri*) is an example of success of an eradication programme. This disease was first noticed in Florida citrus trees in 1913. An eradication campaign was started in 1915. All the citrus nurseries and orchards were inspected and the infected trees were cut and burnt. The eradication programme continued till 1927 and no citrus canker was present in that area. Peach yellows and peach rosette were also controlled by removal and destruction of diseased trees. In Tamil Nadu also there were some eradication campaigns launched under Destructive Pests and Diseases Act. Eradication programme was set up to control bud rot of palms and completed with success. At Sathyamangalam eradication of sandal wood tree affected by spike disease was also made to contain this disease.

iv. Eradication of pathogens from infected plant parts by surgery: Eradication of affected plant parts (tree surgery) are also practiced in certain cases which reduces the source of primary inoculum. Lesions caused by fire blight bacterium

(*Erwinia amylovora*) on pear and apple trees are removed during winter months. This not only prevents further spread in the affected trees but also reduces the amount of inoculum that can spread to other branches and trees. The cankered areas in the branch or trunk of almond and pear trees caused by *Ceratocystis fimbriata* are surgically removed and the trees are saved. Tree surgery is also practiced in coconut trees affected by stem bleeding disease (*Ceratocystis paradoxa*), citrus gummosis (*Phytophthora citrophthora*), *Dendrophthoe* spp. on citrus, bud rot of palms (*Phytophthora palmivora*) and koleroga of arecanut (*P. arecae*)

v. Eradication of culled out plant materials, debris etc. through different cultural practices

2. Crop rotation

Soil borne pathogens that infect plants of one or few species or even families of plants can sometimes be reduced in the soil by planting non-host crops for 3 or 4 years. In this case crop rotation can reduce population of pathogen (e.g. *Verticillium*). In some cases, the field is tilled and left fallow for a year or part of year. During fallowing, pathogen debris and inoculum are destroyed by microorganism with little or no replacement. In areas with hot summer, fallowing allows greater heating and drying of the soil, which leads to a marked reduction of nematodes and some other pathogens. Other cropping systems utilize herbicides, reduced tillage and fallowing. In such systems, certain diseases, e.g. stalk rot of grain sorghum and corn, caused by *Fusarium moniliforme*, have been reduced dramatically. In contrast other diseases, such as Septoria leaf blotch of wheat, and barley scab were increased.

3. Sanitation

Sanitation consists of all activities aimed at eliminating or reducing the amount of inoculum present in a plant, field or a warehouse and at preventing the spread of the pathogen to other healthy plants and plant products. Thus, ploughing under infected plants after harvest, such as leftover infected fruit, tubers or leaves, helps cover the inoculums with soil and speed up its disintegration and concurrent destruction of most pathogens carried in or them. Similarly, removing the infected leaves of house or garden plants helps remove or reduce the inoculum. In some parts of world, infected crop debris of grasses and rice crops is destroyed by burning, which reduces or eliminates the surface inoculum of several pathogens.

By washing their hands before handling certain kinds of plants, such as tomatoes; workers who smoke may reduce the spread of *tobacco mosaic virus*. Disinfecting the knives used to cut propagative stock, such as potato tuber and disinfecting pruning shears between trees reduce the spread of pathogen through such tools. Washing the soil of farm equipment

before moving it from one field to another. Also, it may also help prevent the spread of pathogens present in soil.

4. Creating conditions unfavourable to the pathogen

Stored product should be aerated properly to hasten the drying of their surface and inhibit germination and infection by any fungal or bacterial pathogens present on them. The appropriate choice of fertilizers or soil amendments may also lead to change in the soil pH, which may unfavorably influence the development of pathogen. In the production of many crops, particularly containerized stock, using decomposed tree bark in the planting medium has resulted in the successful control of diseases caused by several soil borne pathogens, e.g. *Phytophthora*, *Phythium* and *Thielaviopsis* causing root rots, *Rhizoctonia* causing damping off and crown rot, *Fusarium* causing wilt, and nematode diseases of several crops.

5. Polyethylene traps and mulches

Many plant viruses, such as cucumber mosaic virus are brought into crops such as peppers, by airborne aphid vectors. When vertical, sticky, yellow polyethylene sheets are erected along edges of susceptible crops fields, a considerable number of aphids are attracted to and stick to them. If reflectant aluminum or black, whitish-grey or colored polyethylene sheets are used as mulches between the plants or rows in the field, incoming aphids, thrips and possibly other insect vectors are repelled and misled away from the field. Reflectant mulches, however, cease to function as soon as the crop canopy covers them.

6. Evasion or avoidance of the pathogen

For several plant diseases, control depends on attempts to evade pathogens. For example, bean anthracnose, caused by the fungus *Colletotrichum lindemuthianum*, and the bacterial blight of bean, caused by bacteria *Xanthomonas phaseoli* and *Pseudomonas phaseolicola*, are transmitted through the seed. They can be successfully controlled by using disease free seed and seed treatments. In many crops, susceptible crop is planted at a great enough distance from field containing possibly diseased plants so that the pathogen would not likely infect the crop. This type of crop isolation is practiced mostly with perennial plants, such as peach orchards isolated from choke cherry shrubs or trees infected with X disease phytoplasma.

In addition, growers carry out numerous activities aimed at helping the host evade the pathogen. Such activities include using vigorous seed, selecting proper dates and proper sites,

maintaining proper distances between fields and between rows and plants, planting wind breaks or trap crops, planting in well drained soil, and using proper insect and weed control. All such practices increase the chances that the host will remain free of pathogen or at least that it will go through its most susceptible stage before the pathogen reaches the host.

7. Use of pathogen free propagating material

When a pathogen is excluded from the propagating material of the host, it is often possible to grow the host free of that pathogen for the rest of its life. Examples are woody plants affected by non-vectored viruses. All types of pathogen can be carried in or on propagating material. True seed, however, is invaded by relatively few pathogens, although several may contaminate its surface. Seed may carry internally one or a few fungi such as those causing anthracnose and smuts, certain bacteria causing bacterial wilts, spots and blights and certain viruses (tobacco ring spot in soybean, bean common mosaic virus, lettuce mosaic, barley stripe mosaic, squash mosaic and prunus necrotic ring).

8. Pathogen free seed

Seed that is free of fungal, bacterial and some viral pathogens is usually obtained by growing the crop and producing seed in (1) an area free of or isolated from the pathogen. (2) an area not suitable for the pathogen (e.g., the arid western regions of the United States where bean seed is produced usually free of anthracnose and bacterial blights). (3) an area not suitable for the vector of pathogen (e.g., the northern or high altitude fields where aphids, the vectors of many viruses are absent).

Testing seed for fungal and bacterial pathogen is done by symptomatology, microscopically and by culturing the pathogen on general or selective media. For testing of bacteria, serological tests are also used. In the 1980's, the sensitivity and accuracy of detection and identification of all types of pathogens was increased greatly by use of polymerase chain reaction.

9. Pathogen free vegetative propagating material

Vegetative propagating material free of pathogen that are distributed systemically through out the plant is obtained from mother plant that had been tested and shown to be free of particular pathogen or pathogens. To ensure continuous production of pathogen free buds, grafts, cuttings, rootstocks and runners of trees, vines, and other perennials, the mother plant is indexed for the particular pathogen at regular intervals.

For certain crops, such as potato, complex certification programs have evolved to produce pathogen free seed potatoes. For the seed to be certified the plants must show disease level no higher than those allowed by particular state. Sometimes it is impossible to find even a single plant of variety that is free of particular pathogen, especially of viruses. In that case, one or few healthy plants are initially obtained by tissue culture of upper millimeter or so of one of the growing meristematic tip of the plant, which most viruses do not invade.

10. Exclusion of pathogen from plant surfaces by epidermal coatings

Successful results at controlling disease of above ground parts of plants have been obtained in experiments in which the plants were sprayed with compounds that form a continuous film or membrane on the plant surface and inhibit contact of pathogen with the host and penetration of host. Such a high quality of lipid membrane forms, for example, when plants were sprayed with water emulsion of dodecyl alcohol. The membrane allows diffusion of oxygen and carbon dioxide but not of water. The membrane is not easily washed by rain and remain intact for about 15 days. Similarly, kaolin based films have proved effective in protecting apple shoot from becoming infected with bacterial disease fire blight and apple fruit from powdery mildew. It also protects grapevine from Pierce's disease caused by *Xylella fastidiosa* by interfering with its transmission by the vector.
